



Research Article

Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*)

Submission ID aede2cc9-345a-4d77-8d67-a6c54ef9d5ae

Submission Version Initial Submission

PDF Generation 16 Sep 2026 16:10:01 EST by Atypon ReX

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Additional Information

Keywords

Mountain pine beetle; Disturbance refugia; Curculionidae; Scolytinae; British Columbia

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

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Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*)

Terrain, stand density and flight-period wind as controls on attack in the Selkirk Mountains of British Columbia

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Keywords. Mountain pine beetle, Disturbance refugia, Curculionidae, Scolytinae, British Columbia

Data availability statement. Public datasets were used in this analysis. Beetle disturbance was classified from Landsat Collection 2 Level-2 surface reflectance, stand structure was derived from British Columbia's provincial Vegetation Resources Inventory, terrain from the Natural Resources Canada High Resolution Digital Elevation Model (Natural Resources Canada, 2017) and wind from Environment and Climate Change Canada hourly station records. All derived data and the complete analysis code that reproduce every number, table and figure in this article are at <https://github.com/seamusrobertmurphy/beetle-topography-and-wind-study> and will be deposited in the Dryad Digital Repository, with a DOI, with the revised manuscript. The study's classifier was trained on the 28 field plots of beetle-killed basal area established in July and August 2020 in the Darkwoods Conservation Area by Murphy, Leslie, Wilson and Banks (Murphy et al., 2026), whose plot locations, killed-basal-area measurements and Landsat classifications of red-stage mortality were used with the permission of those authors; that study and its co-authors, Adrian Leslie, John Wilson and Lauren K. Banks, are acknowledged as the source of the ground truth on which every attack map in this article rests.

Conflict of interest statement. The author has no conflict of interest to declare. The author was the lead author of the earlier study on the same ground (Murphy et al., 2026), and the field plots on which the classifier in this article was trained were collected under that study with its co-authors, Adrian Leslie, John Wilson and Lauren K. Banks. Those data are used here with their agreement, there is no dispute over the ownership of any data presented, and every contribution to the present article has been attributed by authorship or acknowledgement.

Author contributions. Seamus Murphy conceived and designed the present study, assembled the datasets, wrote the analysis code, performed the analysis, prepared the figures and tables, and wrote and revised the manuscript, which is every CRediT role. The 28 field plots of beetle-killed basal area and the Landsat classifications of red-stage mortality were produced under the earlier study by Murphy, Leslie, Wilson and Banks (2026), and the co-authors of that study contributed to the field data collection and the ground-truthing that this article reuses but took no part in the design, analysis or writing of the present article.

Abstract

1. Disturbance refugia from mountain pine beetle (*Dendroctonus ponderosae*) outbreaks have been hypothesised by Krawchuk et al. (2020) to form where tree defences remain effective, on shaded ground that spares trees water stress or in thin stands with few large hosts where wind may disrupt the aggregation pheromone.
2. All three conditions were tested across 5,573 ha of the Selkirk Mountains, British Columbia, using 59 sixteen-day Landsat epochs over eight outbreak years with annualized forest inventory, geomorphometric variables and a terrain-resolved MicroMet wind field from hourly station records.
3. Wind disruption was supported conditionally, in that attack fell where a thin stand and strong flight-period wind coincided, the density-by-wind interaction -0.049 for stem density ($P < 0.001$) and -0.017 for standing volume ($P < 0.05$), both remaining after within-season contagion entered the model.
4. Terrain acted differently. Leeward ground had more attack as a main effect, -0.267, independent of stand density, and open gentle ground more still, sky view +0.285, which is the pattern expected if wind-borne beetles, which land without discriminating hosts, settle where the flow slows. Attack responded to host size as a threshold, peaking at 31.5 per cent in the 25 to 30 cm quadratic mean diameter class.
5. The shade hypothesis failed, since north-facing slopes showed more attack (+0.384). Refugia are therefore proposed to lie on warm, windward ground in thin stands during windy flight periods, and the terrain ruggedness signal of the companion study resolves into shelter and openness.

1. Introduction

Mountain pine beetle (*Dendroctonus ponderosae* Hopkins [Coleoptera: Curculionidae: Scolytinae]) has impacted more lodgepole pine (*Pinus contorta* Douglas ex Loudon) stands across British Columbia than any other disturbance event on record (Sambaraju & Goodsman, 2021; Taylor & Carroll, 2003). However, mortality is not distributed evenly across a landscape, and the stands that survive supply the structure and seed from which the next forest develops. Such stands are termed disturbance refugia, places buffered from disturbance over time (Krawchuk et al., 2020). Locating a refugium does not explain it, and explanation requires a mechanism linking survival to a measurable property of the site, such as terrain, soil or stand structure (Cartwright, 2018). Krawchuk et al. (2020) proposed such a mechanism, that refugia could occur “in areas with cooler temperatures (eg from topographic shading) that protect trees from water stress; in areas with lower host density, allowing for greater wind disruption of beetle pheromone communication and more vigorous tree growth and chemical defenses; and in areas with fewer large-diameter host trees” (p. 239). These are three testable claims. Topographic shading reduces attack by relieving water stress on cool ground. Low host density reduces attack by admitting the wind that disperses the aggregation pheromone. A scarcity of large-diameter hosts reduces attack by limiting brood production, because stems under 25 cm in diameter are sinks for the beetle and stems above it are sources (Carroll & Safranyik, 2004), and attack cannot occur where the host is absent, so a cell without pine is not a refugium (Cartwright, 2018). Two of the three act through terrain. This study fitted the three together on one landscape, following Cartwright (2018), who modelled the controls on an insect refugium in stands of low basal area, and Maher et al. (2021), who tested refugia from this beetle on transects at alpine treeline.

The part wind plays in the beetle’s dispersal is well studied. Within a stand, dispersal follows the wind close to the ground (Safranyik et al., 1989; Safranyik et al., 1992), while above the canopy

beetles have been tracked by radar to more than 800 m and moved tens of kilometres in a day (Ainslie & Jackson, 2010; Chen & Jackson, 2017; Jackson et al., 2008), and the factors that govern dispersal by flight are reviewed by Jones et al. (2019). The wind mechanism of the refugia hypothesis is narrower and concerns conditions inside the canopy during an attack. Cartwright (2018) reasoned that “thinner stands also increase wind penetration, helping to disperse beetle pheromones and disrupt chemical communications needed to coordinate attacks”, and Powell & Bentz (2014) give the converse as a condition for outbreak. Stand density was therefore kept in every model fitted here, since a model that removes it and then reads a terrain coefficient as a wind effect has removed the pathway it set out to test. On the same reasoning, ground exposed to the wind and periods of stronger flight-period wind should both have less attack (Jones et al., 2019; Krawchuk et al., 2020). Two constraints set the interval over which such a wind term can be measured. Flight is confined to a temperature window, between 19 and 41 degrees C, on bright afternoons when “peak flight is in the early to mid-afternoon” (Gray et al., 1972; McCambridge, 1971; Safranyik & Carroll, 2006). A daily or monthly mean wind speed averages across many hours in which no beetle is flying, and radiation during the flight window is a different quantity from radiation accumulated across the season, which is the quantity the shading pathway concerns. Mass attack is also a threshold phenomenon, the irruption threshold being “the population density at which endemic populations may transition towards the epidemic state” (Cooke et al., 2025; Howe et al., 2022). Attack in one year is thus not independent of attack in the year before, which is why previous-year and neighbourhood pressure enter the models, and an environmental variable acts on this system by regulating that threshold rather than by adding to attack. A wind effect through the plume is then expected as an interaction with host density and not as a main effect. Shade and vigour both predict less attack on cool ground by routes this design cannot separate, since cool sites slow development (Sambaraju & Goodsman, 2021) while water stress does not act on defence in one direction (Netherer et al., 2021).

Where a dispersing beetle comes down changes what a terrain main effect can mean. Hynum & Berryman (1980) monitored landing on lodgepole pine with landing traps and found that beetles “were unable to distinguish between hosts, dead hosts and nonhosts during landing”. Where a beetle lands is decided by its transport rather than by the tree beneath it, and a beetle descending from transport above the canopy arrives as a wind-borne particle. de la Giroday et al. (2011) set out what follows, that landscape features “provide impactive surfaces for interception of insects” and that settlement rises “in areas where wind speed is reduced”. The ground where the flow slows, the lee of ridges and sheltered slopes, is where such a beetle should come to rest, which is the pattern the wind shelter index of Plattner et al. (2004) was built to predict for snow. Deposition and plume disruption thus make different predictions, and the difference is what this design can test. Deposition acts before any host is chosen and predicts a main effect of terrain shelter that does not depend on stand density, whereas plume disruption acts on an aggregation already under way and predicts an interaction between stand density and wind with no requirement that shelter act alone. A terrain coefficient read without this distinction is assigned to a mechanism it may not belong to. Landform enters for a different reason, in that infested groups are reported in draws and gullies and deep snow insulates overwintering brood (Safranyik & Carroll, 2006), while elevation enters as a composite of temperature, snowpack, season length and host distribution that this design cannot separate (Sambaraju & Goodsman, 2021).

This study grew out of a companion study of conifer regeneration after the 2015 Mt Midgeley fire on the same ground (Murphy et al., 2026), which classified burn severity and red-stage beetle mortality from Landsat time series against field reference plots, stem-mapped 3,886 seedlings of seven conifer species by sub-metre GPS across three 4 ha plots, and fitted inhomogeneous log-Gaussian Cox process models of seedling intensity to distance from seed source, burn severity, beetle mortality, aspect, wind and terrain ruggedness. The present study took from it the 28 field plots of 20 by 20 m in which beetle-killed basal area was measured against pitch tubes, frass and

gallery architecture, its 30 m Landsat grid and its perimeter. In that model terrain ruggedness was the largest terrain effect on seedling intensity, $+0.626$ ($P < 0.001$). A ruggedness coefficient says that the shape of the ground governed what survived and regenerated without saying which property of that shape the beetle responded to. The present study turned the beetle outbreak that model treated as a covariate into the response and replaced the single ruggedness index with the four terrain properties the beetle's biology names, exposure to the prevailing wind, openness to the sky, position on the slope and depth of the valley, so that the effect of relief could be separated into its parts.

The study addressed three questions. Do the three mechanisms Krawchuk et al. (2020) named, stand density, topographic shading and the scarcity of large hosts, predict moderate-to-high disturbance once host, terrain and previous attack are in the model? Does wind act on attack through stand density, which is the form plume disruption takes, and only where wind varies in time? Does terrain shelter act as a main effect, which is what deposition of wind-borne beetles predicts, or only through stand density, which is what plume disruption predicts?

2. Methods

2.1 Study area

The study area covered 5,573 ha of the Selkirk Mountains in southeastern British Columbia, 61,923 cells of 30 m spanning 830 to 1,744 m, 914 m of relief, on the grid of the parent study, so that results compared directly with it. The perimeter was centred on the 2015 Mt Midgeley fire, the parent study's site, and extended beyond its 480 ha of burned area to take in the range of stand density the pheromone mechanism needed. The extension was constrained rather than arbitrary. The burn was buffered by 5 km and the buffer was then cut to the elevation band of the parent study's site, so that the ground added was comparable to the ground it was added to. Table S1 lists the datasets the study combined and the spatial and temporal resolution of each. Those

resolutions were uneven, and the analysis depended on that unevenness. The response was measured every sixteen days, the inventory once a year and the station winds every hour, and the terrain was measured once.

Figure 1 near here

2.2 Beetle disturbance and flight-windows

The response was red-stage beetle attack in each of eight outbreak years, 2006 to 2011, 2013 and 2014, mapped by the method of Murphy et al. (2026) over the study perimeter. That study mapped red-stage mortality from Landsat surface reflectance as the change in the normalised difference moisture index (NDMI) between the pre-outbreak scene, its pre-disturbance baseline, and one scene in each later year, and validated the map against 28 field plots of 20 by 20 m in which beetle-killed pine was confirmed from pitch tubes, frass and gallery architecture and measured as the fraction of plot basal area killed. NDMI was chosen from four candidate indices, the other three being tasselled-cap wetness, greenness and brightness, because it separated attacked from unattacked plots most cleanly, predicted killed basal area most closely and classified red stage most accurately ($\kappa = 0.750$, overall accuracy 0.824). The full diagnostics are reported there. Scenes were dated 7 to 31 August in every year but 2014, because red needles are fully expressed by the end of the first summer after attack and a near-anniversary date holds leaf phenology and sun angle constant between the two images being differenced. The present study kept that baseline, that index and that season, building each annual image as the median of the cloud-masked Landsat scenes from 1 June to 31 August, with Landsat 5 for 2005 to 2011 and Landsat 8 for 2013 and 2014 rescaled to the Landsat 5 range over forest that showed no attack, and subtracting the pre-outbreak NDMI from each later year. Each year was mapped separately with water masked, and the years were never merged, so that the year-to-year variation the study measured was kept. Annual prevalence inside the perimeter ran from 3.9

to 18.6 per cent, pooled 9.7 per cent over 111,707 cell-years (Table S2), and Figure 2 shows where the outbreak arrived first and how much of the perimeter it reached each year.

[Figure 2 near here](#)

The flight window was 1 July to 15 August and the hours 12:00 to 17:00. Neither bound was chosen from these data, which is what multi-year phenology studies of forest insects do when they fix activity windows from monitoring rather than from the response (Pawson et al., 2021). The dates were the flight period Safranyik & Carroll (2006) give for this region. The hours followed their “peak flight is in the early to mid-afternoon” together with the 11:00 to 14:00 emergence peak of Gray et al. (1972), taken to the later side because emergence precedes the flight it starts.

The window was checked against the station climate. [Figure 3](#) presents time series of 236,079 hourly station records from May to September of the nine study years. Inside the window, 89.5 per cent of afternoon hours fell within the 19 to 41 degrees C flight range, against 51.4 per cent outside it, and the afternoon mean was 26.0 against 19.2 degrees C, so the window roughly doubled the share of hours in which flight was thermally possible. Wind followed the same daily pattern, with mean speed peaking in the same hours as temperature, at 10.4 km/h in mid-afternoon against 4.6 km/h at dawn. The hours in which the beetle could fly were therefore also the windiest of the day.

Station wind from Environment and Climate Change Canada entered hour by hour, and the terrain adjustment described next acted on each observation rather than on a mean. Wind was then summarised by sixteen-day epoch, the interval at which the response was measured and the finest at which wind varied within a season as well as between seasons.

[Figure 3 near here](#)

2.3 Terrain-resolved wind

Station wind interpolated from four to seven valley stations was nearly flat within a year, so wind was also computed as a field varying in space with the MicroMet model of Liston & Elder (2006), whose wind component is seven equations implemented directly from the source paper and set out in supplementary materials ([S1](#)). The terrain weighting factor depends on direction and not on speed. It was computed once for each of 16 wind direction sectors of 22.5 degrees, the sixteen points of the compass, and each hourly observation was multiplied by the surface for its own sector, so that nothing was averaged before the terrain acted on it. Over all sectors the factor ran from 0.60 to 1.38.

Speed and direction were combined as vector components, because averaging degrees across the 360 to 0 discontinuity is not meaningful. The resulting field varied from 1.9 to 2.8 km/h across the grid within a year. It was not a terrain index under another name, correlating +0.148 with the windward-leeward index and +0.123 with flight-window radiation, while its strongest association was with elevation at +0.320.

2.4 Stand structure

Stand structure came from the provincial Vegetation Resources Inventory layer VEG_COMP_LYR_R1_POLY, retrieved within the perimeter. Six of its attributes covered the mechanism and the stem-size threshold, total basal area, crown closure, live stems per hectare, quadratic mean diameter over stems of 12.5 cm and larger, stand age, and susceptible pine basal area, formed as total basal area times the pine share of cover.

Every model was fitted on annualized stand structure. Each cell-year took the inventory snapshot the province published for that year, depleted for harvest and projected for growth to it, so the host terms varied in time as well as in space. One year was substituted, because the 2007 inventory dataset omitted basal area and live stems variables, so the 2006 record stood in for

2007 rather than an average derived from 2006 and 2008 records. The inventory is itself a projection, so an average would produce a stand structure the province never published, whereas the 2006 record is one it did. Cell-years at or above the 25 cm source-sink threshold made up 67.1 per cent of the sample ([Table 1](#)).

[Table 1 near here](#)

The inventory is a projected operational product rather than a census, and its polygons here had reference years spanning several decades. A polygon interpreted from late-outbreak photography described a stand the beetle had already attacked, and basal area and pine cover were post-attack over part of the study window. Unattacked vintages were not reported in this study, and the limitations this presents in the results is taken up in the Discussion.

2.5 Geomorphometry

Terrain was described by surfaces computed with SAGA GIS over the full reprojected elevation model and clipped afterwards, so that a search radius near the boundary still fell on measured ground. The set separated the single ruggedness index of Murphy et al. (2026) into the properties the beetle's biology points to, while ruggedness itself was kept as a candidate so that the separation was tested against it rather than assumed.

Radiation was computed twice because two mechanisms require different quantities. Flight-window radiation was the direct and diffuse total over 1 July to 15 August restricted to 12:00 to 17:00, the hours identified as the flight peak (Safranyik & Carroll, 2006), while growing-season radiation was the whole-day total from 1 May to 30 September, the shading quantity the first mechanism of Krawchuk et al. (2020) concerns. A single annual heat index cannot separate the two, flight-window direct radiation spanning a 7-fold range here against 2.6-fold for the season total.

Exposure entered as the windward-leeward index and effective air flow height at the prevailing bearing, the wind exposition index over all directions, the wind shelter index of Plattner et al. (2004), which is “the maximum gradient within a given radius in upwind direction”, and topographic openness. Shape entered as terrain ruggedness, vector ruggedness, topographic position and its multi-scale form, convergence, slope, curvature and geomorphic class, a landform class read from the pattern of horizons visible around a cell. Landform entered as topographic wetness, valley depth, height above the valley floor, normalised height and mid-slope position, because infested groups are reported in draws and gullies and deep snow insulates overwintering brood (Safranyik & Carroll, 2006), and because slope was one of the two site variables that best explained symptom abundance after attack by the European spruce bark beetle (Kautz et al., 2023). Aspect entered as its northward and eastward components with the heat load index of McCune & Keon (2002).

2.6 Beetle refugia modelling

Candidates were grouped by pathway, stand density, host size, topographic shading, flight-window radiation, terrain exposure to wind, terrain shape, landform and flight-window wind. Selection ran over four stages and was required to keep at least one variable from each pathway, so that a filter could not silently remove a hypothesis the Introduction established. Candidates whose univariate logistic fit was not significant at 0.01 left first. Clusters at an absolute correlation of 0.75 then kept their member with the highest univariate area under the receiver operating characteristic curve (AUC), variables were removed until every variance inflation factor was below 5, and a lasso penalty chosen by ten-fold cross-validation at the one-standard-error rule, the selection Murphy et al. (2026) used, removed the rest, with the highest-ranked survivor of each pathway exempt. Table S3 gives every candidate and the stage at which it left. The 15 variables that entered the models, 8 of them terrain, are listed in [Table 2](#) with the direction expected of each. Fitting used a class-balanced sample of 42,791 cell-years, 4,000 of each class per year, because

landscape prevalence was about 10 per cent and an unbalanced fit at that prevalence would report the magnitude of the intercept rather than the effect of the covariates.

[Table 2 near here](#)

Every model was a logistic regression of moderate-to-high disturbance y_{it} in cell i and year t on standardised covariates,

$$\text{logitPr}(y_{it} = 1) = \alpha + \mathbf{x}_{it}^T \boldsymbol{\beta} + \gamma_{g(i)} + \sum_k \delta_k z_{k,it} \quad (1)$$

where $\gamma_{g(i)}$ was the effect of the geomorphon landform class g of cell i and the $z_{k,it}$ were the interaction terms, so that every coefficient was a change in log-odds per standard deviation of its variable. Four annual models were fitted in sequence, each adding one mechanism to the one before, M0 host size, shading and landform, M1 stand density, M2 terrain shape, terrain exposure and flight-window radiation, and M3 the interactions of stand density with terrain exposure, flight-window radiation and flight-window wind. They were compared on AIC and on predictive error on the fitted probabilities. M0 to M2 answered the first question and M3 the second at the annual scale. The sixteen-day models answered the second question again where wind varied within a season, the scale at which the plume mechanism acts. The sixteen-day response was a fall in NDMI of 0.080 or more against the same sixteen-day epoch of the 2005 pre-outbreak imagery, over nine epochs from May in each of the eight years. It was regressed on the same covariates as the annual models, with the epoch wind of the terrain-resolved field replacing the annual wind terms, together with its interactions with stem density and standing volume. The model was then refitted with the previous epoch's attack entered twice, as the cell's own state and as the share of cells attacked within 90 m, the radius at which previous attack predicted current attack most strongly of those tested (Table S-), so that the interaction was read against within-season spread. The third question compared the coefficient of terrain shelter on its own, from M2 and M3, with

the coefficient of the shelter by density interaction from M3. Deposition predicts the first present and the second absent, whereas plume disruption predicts the second present whether or not the first is.

3. Results

3.1 The three mechanisms

Table 3 near here

Each mechanism improved the fit when it entered. AIC fell by 298 when stand density entered, by a further 322 with terrain and flight-window radiation, and by a further 562 with the interactions (*Table 3*).

Table 4 near here

Stand basal area was the density term the penalty kept. It entered at +0.430 log-odds per standard deviation, an odds ratio of 1.537, so a stand one standard deviation above the mean in basal area had 53.7 per cent higher odds of moderate-to-high disturbance ($P < 0.001$), the direction a canopy that keeps the pheromone plume together implies. Live stems and crown closure left at the penalty stage (*Table S3*), and crown closure is in any case recorded by the inventory to a ceiling of 60 per cent.

Table 5 near here

Host size acted as a threshold rather than a gradient. Attack peaked in the 25 to 30 cm class at 31.5 per cent and fell above it, to 22.4 per cent at 30 to 40 cm and 13.2 per cent above 40 (*Table 5*). The step across the 25 cm source-sink boundary ran from 21.5 to 31.5 per cent, +10.0 percentage points (95 per cent confidence interval 8.9 to 11.1, $P < 0.001$). In the full model diameter was -0.341 per standard deviation ($z = -15.37$), negative because a linear term fitted through a humped response returned the slope of its falling limb, so the class table rather than the coefficient was

the result here. Across all six classes attack depended on diameter class [$\chi^2 = 678.2$ on 5 degrees of freedom, $P < 0.001$, Cramer's $V = 0.126$], while 30 m cells in a spreading outbreak were not independent, so both P-values were anti-conservative.

Topographic shading was not supported. Radiation entered the model once, as direct radiation during the flight window, at +0.319, the term that represents the temperature limit on flight, while growing-season radiation did not survive selection, so northness, which correlated -0.824 with it, stood in for the shading pathway. Northness entered at +0.384, so shaded, north-facing ground had more attack rather than less, against the prediction.

3.2 Wind through stand density in time

Stand density interacted negatively with terrain-resolved wind, the form the pheromone mechanism predicted. Attack fell where a thin stand and strong wind coincided, the interaction being -0.049 for stem density ($P < 0.001$) and -0.017 for standing volume ($P < 0.05$). Expressed as odds, each standard deviation of the epoch wind regime multiplied the contribution of stem density by 0.952, a reduction of 4.8 per cent. Both interactions remained after the previous epoch of the same season was entered, so neither was the outbreak's own spread appearing as a wind coefficient.

Wind alone gave no protection. Its main effect was +0.022, marginally more attack rather than less, and the density terms were positive, so a thin stand in still air showed no reduction either and the mechanism appeared only where the two coincided. The interaction was refitted under four definitions of the wind window, and the stem-density interaction was negative under all four and distinguishable from zero under three (Table S4).

Figure 4 near here

Attack recurred where it had already occurred, by a margin large enough to change how every other coefficient should be read. A cell attacked in one year was between 9 and 88 times more

likely to be attacked in the next. Entering persistence and 90 m neighbourhood spread separately, following the autologistic design used for this province, gave +0.725 and +0.692 and raised discrimination from 0.774 to 0.857.

The result that mattered was which environmental terms remained after that term entered. Sky view moved from +0.317 to +0.225, and the wind shelter index from -0.260 to -0.215, both describing conditions a cell had whether the beetle was ever present. Conversely, the shape terms were less stable, convergence moving from -0.081 to -0.045 and profile curvature from -0.064 to -0.036. Part of what the landform terms measured was thus where the outbreak had already been, the same failure the terrain-wind index showed against radiation.

3.3 Terrain shelter alone or through density

Terrain predicted attack after stand structure and radiation were in the model, through shelter and openness rather than ruggedness, since neither terrain ruggedness nor the windward-leeward index survived selection (Table S3). The wind shelter index as computed here rose on slopes that faced the prevailing wind, correlating +0.71 with that orientation, and it entered at -0.267 ($P < 0.001$), so leeward ground had more attack and windward ground less. Sky view factor, the share of the sky visible from a cell, which is high on gentle, open, upper ground, entered at +0.285 ($P < 0.001$), so open, sheltered ground had the most attack. The two valley terms, valley depth and height above the valley floor, left at the penalty stage once sky view and elevation were in the model. Elevation entered at +0.381. Of the shape terms, convergence entered at -0.077 and profile curvature at -0.059, while topographic position was not distinguishable from zero ($P = 0.521$). The landform classes are in [Table 6](#).

[Table 6 near here](#)

The third question was tested by fitting the full model twice. Without flight-window radiation, stand density interacted with terrain shelter at +0.055, which on its own would read as plume disruption

acting through terrain. With radiation in the model, that interaction shrank to +0.023 and was no longer distinguishable from zero ($P = 0.371$), while density interacted with radiation at +0.073. On a range whose prevailing bearing was 258 degrees the windward slopes faced west, and those were also the slopes that took the afternoon sun, so 59 per cent of the apparent shelter interaction was radiation and the remainder was too small to support a claim. The shelter coefficient itself did not move between the two fits. Shelter therefore acted alone, as deposition predicts, and not through stand density, as plume disruption would require.

4. Discussion

4.1 Two scales

The three questions resolved into one pattern. Terrain acted on attack as a main effect of shelter and openness that did not depend on stand density, while wind acted on attack as an interaction with stand density that appeared only where wind varied in time. Deposition predicted the first and plume disruption the second, and neither mechanism produced the other's signature. The one place the two could have been confused, the density by shelter interaction, fell by 59 per cent when flight-window radiation entered and could not then be distinguished from zero, while the shelter main effect did not move. A terrain index alone would therefore have supported the plume mechanism on evidence that was partly sunlight.

The wind half of that pattern remained after the previous epoch of the same season was entered, so it was not the outbreak's own spread appearing as a wind coefficient. Two limits remained attached to it. The interaction was small, roughly a tenth of the stand basal area main effect, and the wind field was terrain-modified station data rather than measurement on the ridge, so what was established was that the modelled wind field behaved as the mechanism required, not that the air itself did.

4.2 What was supported

Of the three mechanisms Krawchuk et al. (2020) proposed, two were supported and one failed. Low host density admitting the wind that disperses the aggregation pheromone was supported in the conditional form the mechanism specified rather than as a main effect. The claim was never that dense stands were attacked more, but that a dense canopy kept a plume together and that the advantage this conferred on the beetle should diminish as wind speed rose, and both halves of that prediction were present. A scarcity of large-diameter hosts was supported as a threshold rather than as a gradient, attack peaking at 31.5 per cent in the 25 to 30 cm class, at the source-sink boundary of Carroll & Safranyik (2004) and falling away on both sides.

Topographic shading was not supported, and it failed on a surrogate rather than on the quantity the mechanism named. Growing-season radiation did not survive variable selection, so northness stood in for the pathway, the two correlating -0.824. Northness was +0.384, so shaded ground had more attack where the mechanism required less. Northness tested aspect, and aspect measured more than shade, which made this the least secure of the three verdicts, while the water-stress pathway Krawchuk et al. (2020) described was not measured here at all.

4.3 Landing zones

Read with the dispersal literature, the terrain results described where beetles came down rather than where they chose to attack. A beetle descending through slowing air behaved as any wind-borne particle did and settled where the flow decelerated, in the lee of ridges and on sheltered slopes, which was what the shelter index measured and why the same index predicted where snow accumulated (Plattner et al., 2004). Three results fitted that reading and none contradicted it. Leeward ground had more attack as a main effect, -0.267, and the effect did not detectably depend on stand density. Open, gently sloping ground had more attack, sky view +0.285, and the valley terms did not survive the penalty beside it. The density by wind interaction that plume

disruption predicted appeared only in the sixteen-day models, where wind varied in time, and not through terrain, where it varied in space. Terrain shelter therefore read as a landing zone and the sixteen-day wind as plume disruption, two mechanisms at two scales rather than one.

A prior study that read terrain this way found the opposite association. de la Giroday et al. (2011) associated the highest infestation intensities in the Peace River region with southwest-facing open slopes and mid-slope ridges, the surfaces that met a westerly wind, after first establishment in canyons and valleys, whereas here the association was leeward and with open ground. Impaction on faces that met the wind and settlement where it slowed were both deposition, and which one a landscape showed may have depended on how much of its relief a descending beetle cleared.

The reading was an interpretation rather than an observation. No beetle was tracked to the ground here, sky view factor also set diffuse radiation, so part of its coefficient may have been sunlight, and open upper ground warmed first, so the same pattern would have followed from faster development as readily as from deposition (Sambaraju & Goodsman, 2021). The component reading was nonetheless more than a ruggedness index could give, since that index recorded that relief mattered without saying how.

4.4 Two measurement lessons

Two measurement results extended beyond this landscape. The first was that a terrain wind index was in part a measurement of incident radiation, flight-window radiation having moved the density by shelter interaction from +0.055 to +0.023. That confusion was unlikely to be peculiar to the present dataset. A shelter index and an afternoon radiation surface were both functions of slope and aspect, and on a range whose prevailing flight-window bearing was 258 degrees the slopes that met the wind were the west-facing ones, which were also the slopes taking afternoon sun during the flight peak. Any study entering a terrain wind index without a radiation term on the same clock would attribute radiation to wind.

The second was that a landform variable in a spreading outbreak recorded in part where the outbreak had already been, previous-year attack having moved convergence from -0.081 to -0.045 while sky view moved only from +0.317 to +0.225. The terms that remained described conditions a cell had whether or not the beetle was ever present, and the terms that moved described its shape, so part of what a landform coefficient measured in this outbreak was the outbreak's own history.

4.5 Limits and management

Elevation was among the largest terms in the model at +0.381, but it was not a single quantity. It combined temperature, snowpack, growing-season length and the distribution of lodgepole pine in a way this design could not separate, so reporting the elevation coefficient as a result would have been reporting a composite.

Three limits lay outside the model and bounded every result in it. The inventory postdated part of the outbreak, since polygons interpreted after the beetle passed described the stand it left, and total basal area discriminated only weakly on its own, a univariate AUC of 0.601. The response was classified rather than observed, from 38 plots inside one burn, so the reported accuracy measured how well the classifier reproduced those plots and not agreement with ground mortality elsewhere. Every conclusion rested on one mountain range across eight years.

For pest management the result meant that refugia could not be mapped from terrain alone, and that the map had two layers. A stand's exposure to the prevailing wind set how many beetles arrived, and its density during the windy weeks of the flight period set how many of those succeeded. Thinning could therefore be expected to lower attack on windward ground during windy flight periods and to do little on the sheltered lee slopes where beetles came down, and only by the modest margin the interaction measured.

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569 **Acknowledgements**

570 This work used no external funding. Beetle disturbance was classified from Landsat Collection 2
571 Level-2 surface reflectance distributed by the United States Geological Survey; stand structure
572 from the British Columbia Data Catalogue; terrain from Natural Resources Canada; and wind from
573 Environment and Climate Change Canada. The author thanks those agencies for maintaining the
574 open archives the study rests on.

575

576 **Tables**

Table 1: Stand structure across the study perimeter, from the Vegetation Resources Inventory, over 111,707 cell-years. SD was the standard deviation of the landscape and SE the standard error of the mean. Skew and Kurt. were the bias-corrected skewness and excess kurtosis.

Attribute	Mean	SD	SE#	Median	Min	Max	Skew	Kurt.
Stand basal area (m ² ha ⁻¹)	35.56	11.60	0.035	37.39	0.98	62.43	-1.01	+1.42
Crown closure (%)	50.10	13.54	0.041	50.00	3.00	70.00	-1.74	+3.37
Live stems (n/ha)	772.75	312.80	0.936	775.00	23.00	4600.00	+1.70	+19.51
Quadratic mean diameter (cm)	27.38	6.40	0.019	26.95	13.55	58.90	+0.73	+1.24
Stand age (years)	115.46	20.88	0.062	116.00	22.00	237.00	-0.78	+1.97
Stand height (m)	26.97	6.27	0.019	27.90	7.00	40.30	-0.26	+0.23
Standing volume (m ³ ha ⁻¹)	272.33	131.38	0.393	281.47	0.82	586.61	+0.01	-0.35
Lodgepole pine cover (%)	20.11	23.93	0.072	10.00	0.00	100.00	+1.45	+1.56
Susceptible pine BA (m ² ha ⁻¹)	7.07	8.85	0.026	4.00	0.00	48.30	+1.63	+2.63

Note the standard error is small because it is computed over 111,707 cell-years, and it measures the precision of the landscape mean, rather than of any one cell.

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*Table 2: The variables that entered the models after selection, the pathway each served, and the direction expected of it from the mechanism named in the Introduction. AUC was the area under the receiver operating characteristic curve of the univariate fit, with the asterisks beside it marking its significance, * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.*

Variable	Pathway	Expected	Univariate AUC
Elevation (m)	Landform	uncertain	0.682****
Susceptible pine BA ($\text{m}^2 \text{ha}^{-1}$)	Host size	positive	0.678****
Stand basal area ($\text{m}^2 \text{ha}^{-1}$)	Stand density	positive	0.601****
Sky view factor	Topographic shading	positive, as main effect	0.686****
Stand age (years)	Host size	positive	0.571****
July mean wind (km/h)	Flight-window wind, stations	negative	0.563****
Northness	Topographic shading	negative	0.586****
Quadratic mean diameter (cm)	Host size	positive above 25 cm	0.510****
June mean wind (km/h)	Flight-window wind, stations	negative	0.534****
Wind shelter index	Terrain exposure to wind	negative, the index rising windward	0.526****
MicroMet flight-window wind (km/h)	Flight-window wind, terrain-resolved	negative	0.580****
Topographic position index	Terrain shape	more attack in convergent terrain	0.585****
Flight-window direct radiation (kWh/m^2)	Flight-window radiation	positive	0.561****
Convergence index	Terrain shape	more attack in convergent terrain	0.530****
Profile curvature	Terrain shape	more attack in convergent terrain	0.514****

Table 3: Comparison of the four annual models, each adding one mechanism to the one before. RMSE was the root mean squared error and MAE the mean absolute error, both on the fitted probabilities, and AUC the area under the receiver operating characteristic curve. Brier skill was the improvement over predicting the prevalence for every cell, where 0 was no better than that base rate.

Model	AIC	ΔAIC	RMSE	MAE	AUC	Brier skill
M0 host size, shading, terrain	41,269	1,182	0.396	0.315	0.758	0.166
M1 + stand density	40,971	884	0.394	0.312	0.761	0.175
M2 + terrain, flight radiation	40,649	562	0.393	0.309	0.767	0.182
M3 + interactions	40,087	0	0.390	0.305	0.778	0.193

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*Table 4: Coefficients of the full annual model M3, continuous terms ordered by absolute size. Each coefficient was the change in log-odds per standard deviation of its variable, fitted on a class-balanced sample, so the intercept was not the landscape prevalence. SE was the standard error and z the Wald statistic. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. The geomorphon landform classes of the same model were reported in Table 6.*

Term	Beta	SE	z
Stand basal area ($\text{m}^2 \text{ ha}^{-1}$)	+0.430****	0.024	18.20
Northness	+0.384****	0.015	24.85
Elevation (m)	+0.381****	0.016	24.18
Susceptible pine BA ($\text{m}^2 \text{ ha}^{-1}$)	+0.343****	0.015	23.65
Quadratic mean diameter (cm)	-0.341****	0.022	-15.37
Flight-window direct radiation (kWh/m^2)	+0.319****	0.023	13.98
July mean wind (km/h)	+0.286****	0.014	20.95
Sky view factor	+0.285****	0.023	12.60
Wind shelter index	-0.267****	0.021	-12.48
Stand age (years)	+0.201****	0.018	11.31
Convergence index	-0.077****	0.015	-5.07
Stand basal area x July mean wind	-0.077****	0.016	-4.81
Stand basal area x Flight-window direct radiation	+0.073***	0.020	3.58
Profile curvature	-0.059****	0.014	-4.10
June mean wind (km/h)	+0.030*	0.014	2.20
Stand basal area x Wind shelter index	+0.023	0.025	0.89
Topographic position index	-0.015	0.023	-0.64

Table 5: Moderate-to-high beetle disturbance by quadratic mean diameter class, on the balanced sample. The 25 cm boundary was the source-sink threshold of the species' bionomics. Intervals were Wilson score intervals on the class proportion, which is why they were asymmetric in the smallest class. Because 30 m cells in a spreading outbreak were not independent, the tests reported in the text were anti-conservative.

QMD class (cm)	n	Attacked	Attacked (%)	95% CI (%)
<15	395	111	28.1	23.9-32.7
15-20	4,101	783	19.1	17.9-20.3
20-25	9,286	1,998	21.5	20.7-22.4
25-30	17,119	5,391	31.5	30.8-32.2
30-40	10,247	2,291	22.4	21.6-23.2
>40	1,643	217	13.2	11.7-14.9

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*Table 6: Geomorphon landform classes in the full annual model M3, as coefficients against the reference class, slope, the most common class on the perimeter. Cells was the number of perimeter cells in each class and Attacked the share of those cells classed moderate-to-high over all years. Flat and pit did not occur on the perimeter. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.*

Class	Cells	Attacked (%)	Beta	SE
slope (reference)	72,262	9.6		
spur	22,050	12.6	-0.075*	0.035
hollow	12,988	6.0	-0.091	0.053
valley	2,963	3.8	-0.676****	0.121
ridge#	1,276	11.0	-1.160****	0.119
peak	168	22.6	-0.523*	0.227

Ridge includes the 16 shoulder cells on the perimeter. No shoulder cell was classed as attacked in any year, so the class could not be estimated on its own and was counted with ridge, its neighbour on the geomorphon scale.

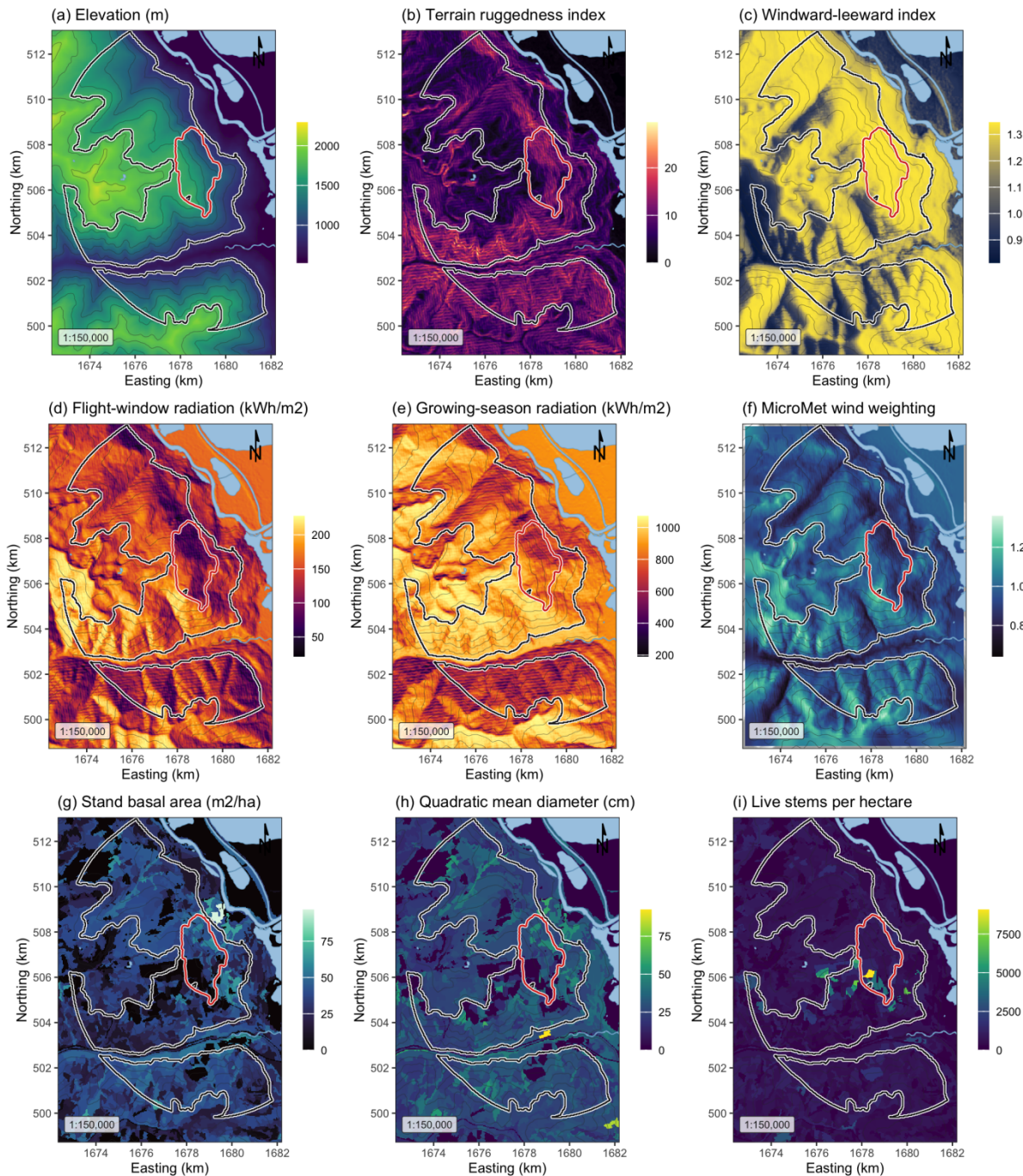
583 **Figures**

Figure 1: Landscape, terrain and stand surfaces across the study area, all EPSG:3153 at 30 m over Esri World Shaded Relief. (a) elevation; (b) terrain ruggedness index; (c) windward-leeward index at the prevailing bearing; (d) flight-window direct radiation, the thermal limit on flight; (e) growing-season direct radiation, the shading pathway; (f) the MicroMet wind weighting factor at the prevailing bearing; (g) stand basal area; (h) quadratic mean diameter, whose 25 cm source-sink threshold fell near the midpoint of the scale; (i) live stems per hectare. The white outline marked the study perimeter and the red outline the 2015 Mt Midgeley burn, with contours at 200 m.

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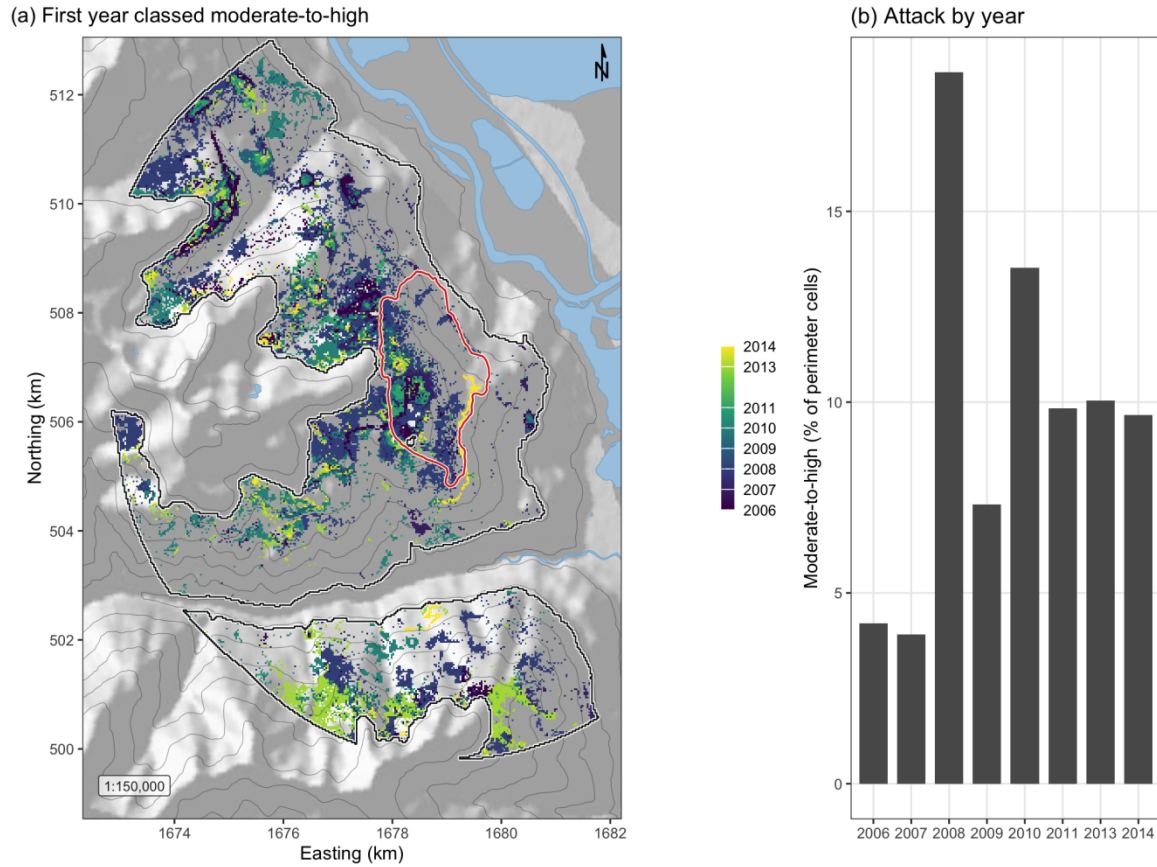


Figure 2: Spread of moderate-to-high beetle disturbance across the study perimeter. (a) The first year in which each cell entered the moderate-to-high class, over shaded relief with the perimeter in white and the 2015 Mt Midgeley burn in red; cells never classed as attacked showed the relief alone. The panel was built from the eight annual maps for display, and those maps were fitted separately and never merged for analysis. (b) The share of perimeter cells classed moderate-to-high in each year.

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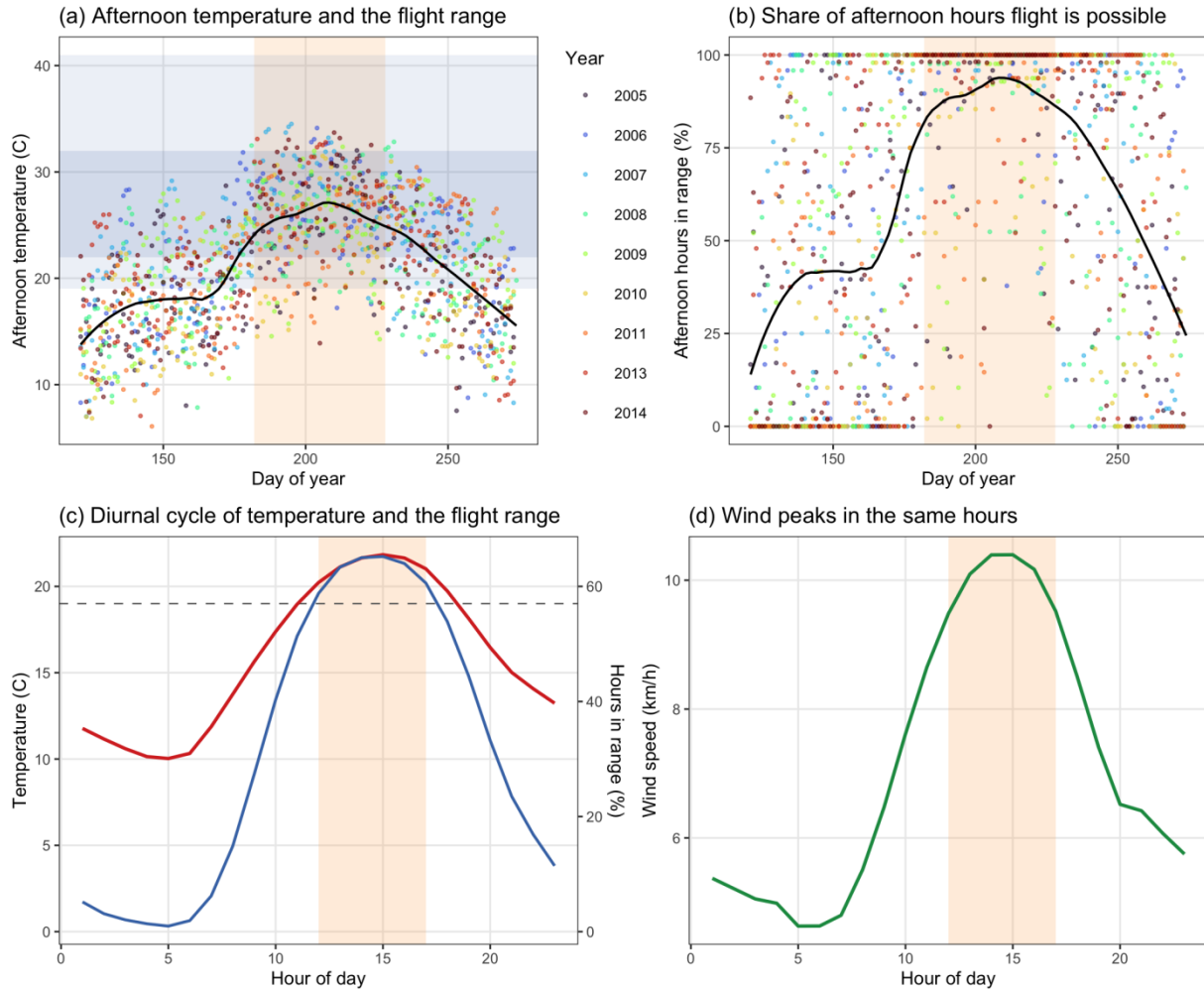


Figure 3: The flight window against the landscape's own climate, from hourly Environment and Climate Change Canada station records for May to September of the nine study years. The shaded band marked the window, 1 July to 15 August, taken from the bionomics and not fitted to these data. (a) Mean afternoon temperature by day of year, with the 19 to 41 degrees C flight range and the 22 to 32 degrees C peak band marked. (b) The share of afternoon hours inside the flight range, day by day. (c) Mean temperature and the share of all hours inside the flight range, by hour, with the 12:00 to 17:00 restriction shaded. (d) Mean wind speed by hour on the same axis, showing that the hours in which the beetle could fly were also the windiest of the day.

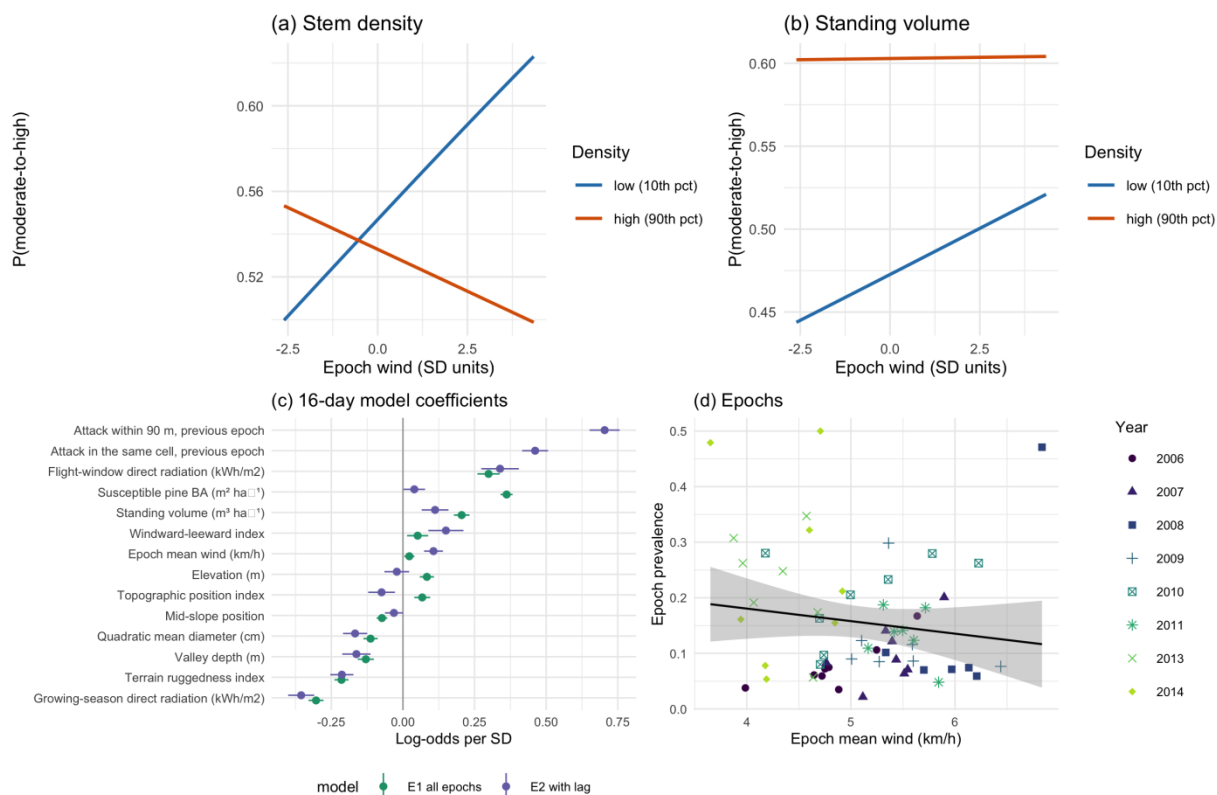


Figure 4: The refugia mechanism as fitted in the sixteen-day models. (a) Predicted probability of moderate-to-high disturbance against terrain-adjusted epoch wind at the 10th and 90th percentiles of stem density, with all other terms held at their means. The lines crossed, so wind raised attack in thin stands and lowered it in dense ones, which was the interaction. (b) The same for standing volume, where the interaction was a fifth the size and flattened the dense-stand line without reversing it, so the lines converged rather than crossed over the observed range. (c) Coefficients of the sixteen-day model with and without the within-season spread term, where the shift of a term between the two fits was the part of it that the spread term absorbed. (d) Prevalence in each epoch against its mean wind, one point per epoch, with years distinguished by shape and shade so that the panel remained legible in black and white.

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Supporting Information

Table S1

The datasets this study combined, with the structure and resolution of each. Spatial resolution was the grid on which a variable was analysed and temporal resolution the interval at which it varied. The response varied every 16 days, the inventory once a year, the station wind every hour and the terrain not at all, which was the unevenness the wind analysis depended on.

Dataset	Source	Variables	Spatial	Temporal	Period	n
Beetle attack (Annual)	Landsat 5 and 8 Collection 2 Level-2	Moderate-to-high NDMI binary	30 m	1 year	2006-2014, (excl. 2012)	8 years
Beetle attack (/16-day)	Landsat 5 and 8 Collection 2 Level-2	Moderate-to-high NDMI binary	30 m	16 days	2006-2014, (excl. 2012)	59 epochs
Stand structure	VRI Historical, BC Data Catalogue	BA, volume, stems, quadratic mean diameter, age, height	30 m (rasterised)	1 year, projected-yr	2005-2014	9 windows
Terrain	NRCan High-Res DEM, SAGA indices	Geomorphons (incl. radiation, exposure, landform)	30 m	Static	n/a	8 fitted
Station wind	Env. & Climate Change Canada	Speed and direction	4 to 7 stations	1 hour	2005-2014, May-Sept	236,079 hourly records
Terrain-resolved wind	DEM-conditioned MicroMet wind field of station data	Weighting factor, modified speed, diverted direction	30 m	16 days, & 1 year	2005-2014	16 sectors (22.5°)
Model frame (Annual)	Rows joined above, (one row / cell-year)	Response & covariates, one row per cell-year	30 m	1 year	2006-2014, (excl. 2012)	111,707 cell-years
Model frame (/16-day)	Rows joined above per Landsat pass (one row / cell-epoch)	Response & covariates, one row per cell-epoch	30 m	16 days	2006-2014, (excl. 2012)	66,302 cell-epochs

Table S2

Moderate-to-high beetle disturbance by year inside the study perimeter.

Year	Cells	Moderate-to-high (%)
2006	13,224	4.2
2007	13,225	3.9
2008	13,285	18.6
2009	13,556	7.3
2010	13,538	13.5
2011	14,944	9.8
2013	14,943	10.0
2014	14,992	9.7

597 **Methods S1**

598 The wind component of the MicroMet model of Liston & Elder (2006) is seven equations,
599 implemented directly from the source paper with its equation numbers given below.

600 Terrain slope β and slope azimuth ξ came from the elevation model, equations 12 and 13 of Liston
601 & Elder (2006). Curvature Ω_c is a cell's elevation minus the mean of the two opposite cells one
602 curvature length scale away, taken on four direction lines and averaged, their equation 14, and
603 the length scale was estimated as the first lag at which elevation autocorrelation fell below 0.5,
604 which was 600 m here. Assigning θ to wind bearing, slope in the wind direction was computed as
605 follows

$$606 \qquad \qquad \qquad \Omega_s = \beta \cos(\theta - \xi) \qquad (1)$$

607 with Ω_s and Ω_c each scaled to $[-0.5, 0.5]$. The terrain weighting factor applied to the observed
608 speed W was derived using

$$609 \qquad \qquad \qquad W_w = 1 + \gamma_s \Omega_s + \gamma_c \Omega_c, \quad \gamma_s = \gamma_c = 0.5 \qquad (2)$$

610 This provided the basis for calculation of the terrain-modified speed using

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614 By [Equation 2](#), W_w depends on direction and not on speed, so it was computed once for each of
615 16 wind direction sectors of 22.5 degrees, the sixteen points of the compass, and each hourly
616 observation was multiplied by the surface for its own sector, so that nothing was averaged before
617 the terrain acted on it, and over all sectors W_w ran from 0.60 to 1.38.

618 **Table S3**

619 Variable selection. Every candidate with its pathway, its univariate AUC, the stage at which it left,
 620 and for the survivors of the inflation stage the lasso coefficient at the chosen penalty. Protected
 621 marked the highest-ranked survivor of each pathway, which the penalty could not remove. AUC
 622 was the area under the receiver operating characteristic curve of the univariate fit, and the
 623 asterisks beside it marked the significance of that fit, * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** P
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Candidate	Pathway	Univariate AUC	Stage	Lasso coefficient	Protected
Elevation (m)	Landform	0.682****	retained	0.336	yes
Susceptible pine BA ($\text{m}^2 \text{ha}^{-1}$)	Host size	0.678****	retained	0.427	yes
Stand basal area ($\text{m}^2 \text{ha}^{-1}$)	Stand density	0.601****	retained	0.347	yes
Sky view factor	Topographic shading	0.686****	retained	0.315	yes
Stand age (years)	Host size	0.571****	retained	0.068	
July mean wind (km/h)	Flight-window wind, stations	0.563****	retained	0.249	yes
Northness	Topographic shading	0.586****	retained	0.314	
Quadratic mean diameter (cm)	Host size	0.510****	retained	-0.124	
June mean wind (km/h)	Flight-window wind, stations	0.534****	retained	-0.025	
Wind shelter index	Terrain exposure to wind	0.526****	retained	-0.205	yes
MicroMet flight-window wind (km/h)	Flight-window wind, terrain-resolved	0.580****	retained	0.253	yes
Topographic position index	Terrain shape	0.585****	retained	-0.168	yes
Flight-window direct radiation (kWh/m^2)	Flight-window radiation	0.561****	retained	0.206	yes
Convergence index	Terrain shape	0.530****	retained	-0.016	
Profile curvature	Terrain shape	0.514****	retained	-0.010	
Height above valley floor (m)	Landform	0.646****	penalty	0.000	
Live stems (n/ha)	Stand density	0.602****	penalty	0.000	
Valley depth (m)	Landform	0.617****	penalty	0.000	
Mid-slope position	Landform	0.517****	penalty	0.000	
Crown closure (%)	Stand density	0.524****	penalty	0.000	
Vector ruggedness measure	Terrain shape	0.600****	penalty	0.000	

Candidate	Pathway	Univariate AUC	Stage	Lasso coefficient	Protected
Terrain ruggedness index	Terrain shape	0.648****	inflation	NA	
Normalised height	Landform	0.584****	inflation	NA	
Flight-window mean wind (km/h)	Flight-window wind, stations	0.558****	inflation	NA	
Eastness	Topographic shading	0.549****	inflation	NA	
Effective air flow height	Terrain exposure to wind	0.682****	collinearity	NA	
Lodgepole pine cover (%)	Host size	0.656****	collinearity	NA	
Slope (degrees)	Terrain shape	0.645****	collinearity	NA	
Positive openness	Terrain exposure to wind	0.636****	collinearity	NA	
Wind exposition index	Terrain exposure to wind	0.563****	collinearity	NA	
Standing volume (m ³ ha ⁻¹)	Stand density	0.555****	collinearity	NA	
Flight-window windy hours (share above 15 km/h)	Flight-window wind, stations	0.548****	collinearity	NA	
Flight-window calm hours (share below 5 km/h)	Flight-window wind, stations	0.547****	collinearity	NA	
Multi-scale topographic position	Terrain shape	0.540****	collinearity	NA	
August mean wind (km/h)	Flight-window wind, stations	0.534****	collinearity	NA	
Plan curvature	Terrain shape	0.528****	collinearity	NA	
solar_flight_diffuse	Flight-window radiation	0.528****	collinearity	NA	
Windward-leeward index	Terrain exposure to wind	0.522****	collinearity	NA	
Growing-season total radiation (kWh/m ²)	Topographic shading	0.517****	collinearity	NA	
Growing-season direct radiation (kWh/m ²)	Topographic shading	0.516****	collinearity	NA	
Flight-window 95th percentile wind (km/h)	Flight-window wind, stations	0.527	univariate	NA	
Stand height (m)	Host size	0.507	univariate	NA	
Topographic wetness index	Landform	0.506*	univariate	NA	
Negative openness	Terrain exposure to wind	0.501	univariate	NA	
Heat load index	Topographic shading	0.499	univariate	NA	

625 Table S4

626 The wind variable of the sixteen-day models was the epoch wind regime, the mean over every
627 hourly station observation in the sixteen days an epoch covered, each hour modified over the
628 terrain by its own bearing. It was not restricted to the flight window, and that choice followed from

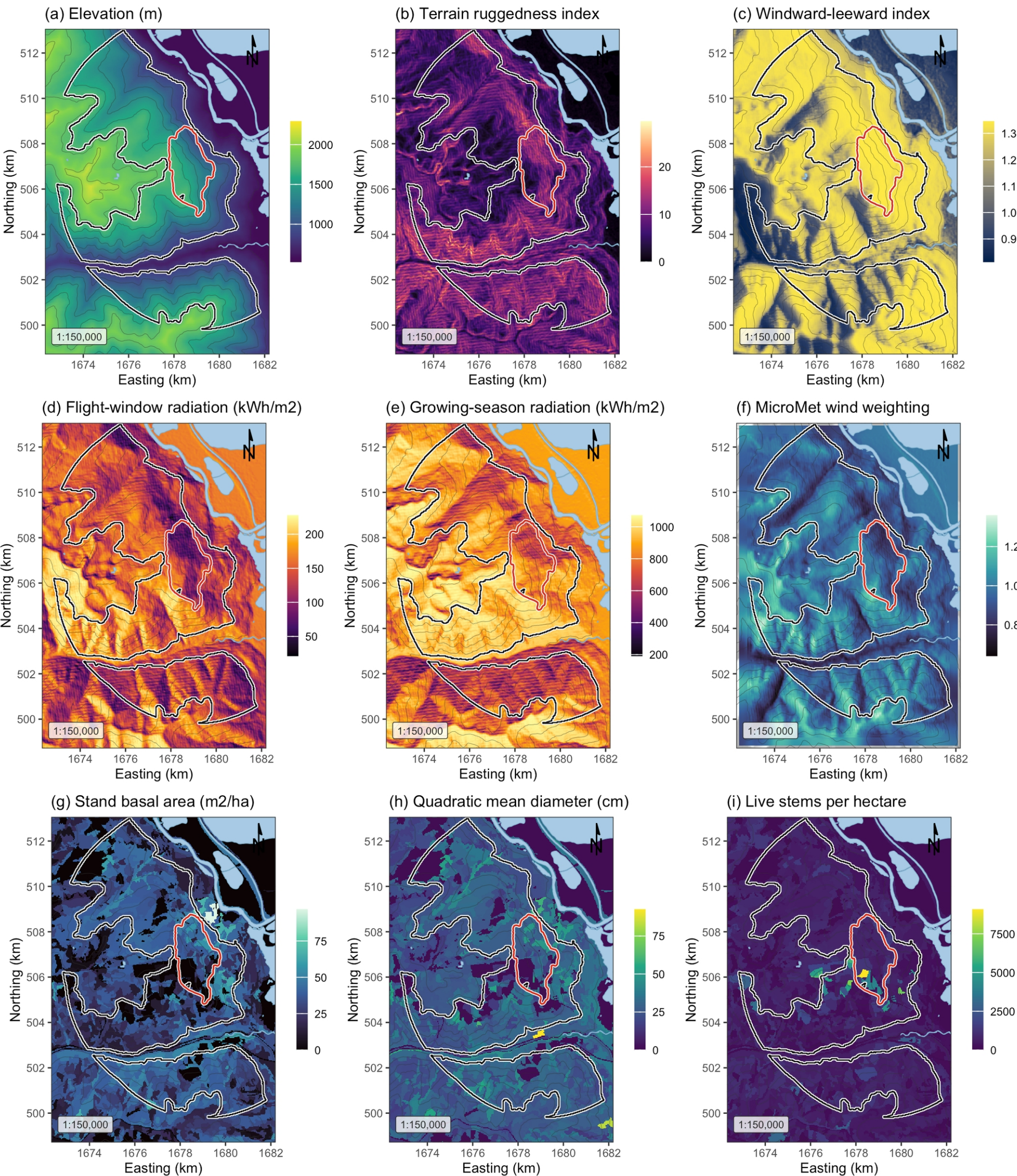
the way the insect was known to disperse. Jackson et al. (2008) tracked mountain pine beetle on weather radar and confirmed the returns by aerial capture, finding beetles “at altitudes up to more than 800 m above the forest canopy” and estimating that those in flight above the canopy “may move 30 to 110 km · day⁻¹”, at a mean density of 4,950 and a maximum of 18,600 beetles per hectare. Ainslie & Jackson (2010) subsequently reproduced that above-canopy dispersion with an atmospheric model. An insect carried in the boundary layer at those heights and over those distances was exposed to the atmospheric regime of the period rather than to the wind measured within a stand between noon and five, and the variable was named accordingly.

The 16-day model was refitted under four definitions of the window (Table S4), every hour, the 12:00 to 17:00 hours the flight peak occupied, the 11:00 to 18:00 hours that fell inside the thermal limits on this landscape more than half the time, and only those hours whose observed temperature fell inside the 19 to 41 degrees C range itself. The stem-density interaction was negative under all four definitions and distinguishable from zero under three, at -0.049 over the full regime and -0.037 and -0.041 over the two clock windows, and fell to -0.011 ($P = 0.173$) when the hours were restricted to those inside the thermal limits, where the standing-volume interaction was instead the larger of the two at -0.040 ($P < 0.001$). One of the two density terms therefore carried the interaction under every definition, but which one depended on how the window was drawn, so the mechanism did not rest on the choice while its expression through stems or through volume did.

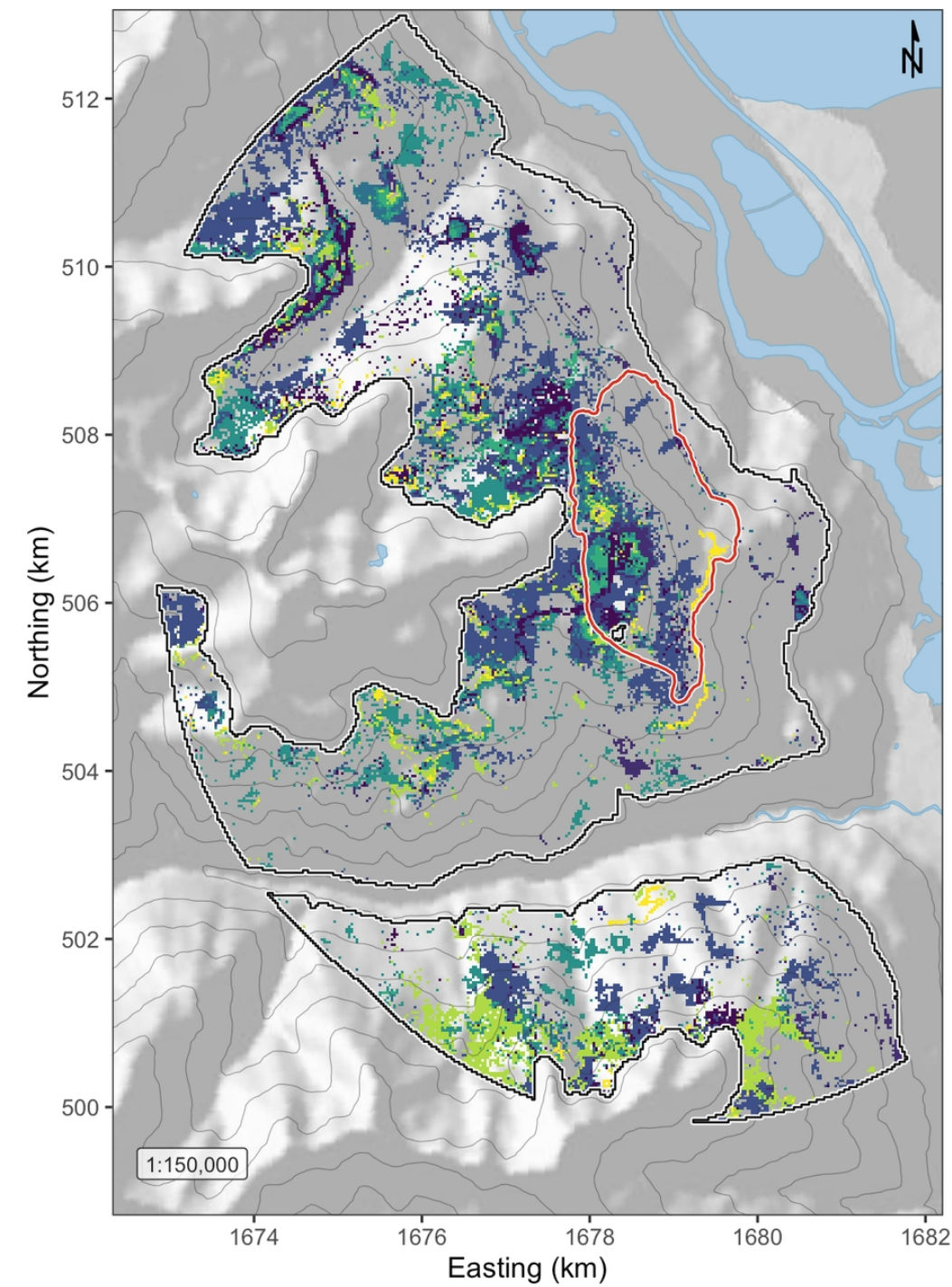
The 16-day model refitted under four definitions of the wind window. The epoch wind regime, every hour, was the definition used throughout the article. Coefficients were log-odds per standard deviation and AUC the area under the receiver operating characteristic curve. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. The thermal-limits row was fitted on fewer cell-epochs because two epochs had too few qualifying hours, so its row was not directly comparable with the other three.

Definition	Cell-epochs	Wind (km/h)	Stems x wind	Volume x wind	AUC
Epoch wind regime (all hours)	66,302	5.01	-0.0491****	-0.0169*	0.673
12:00-17:00	66,302	6.41	-0.0369****	-0.0131	0.673
11:00-18:00	66,302	6.28	-0.0413****	-0.0173*	0.673
Inside the 19-41 C range	64,895	5.64	-0.0108	-0.0397****	0.671

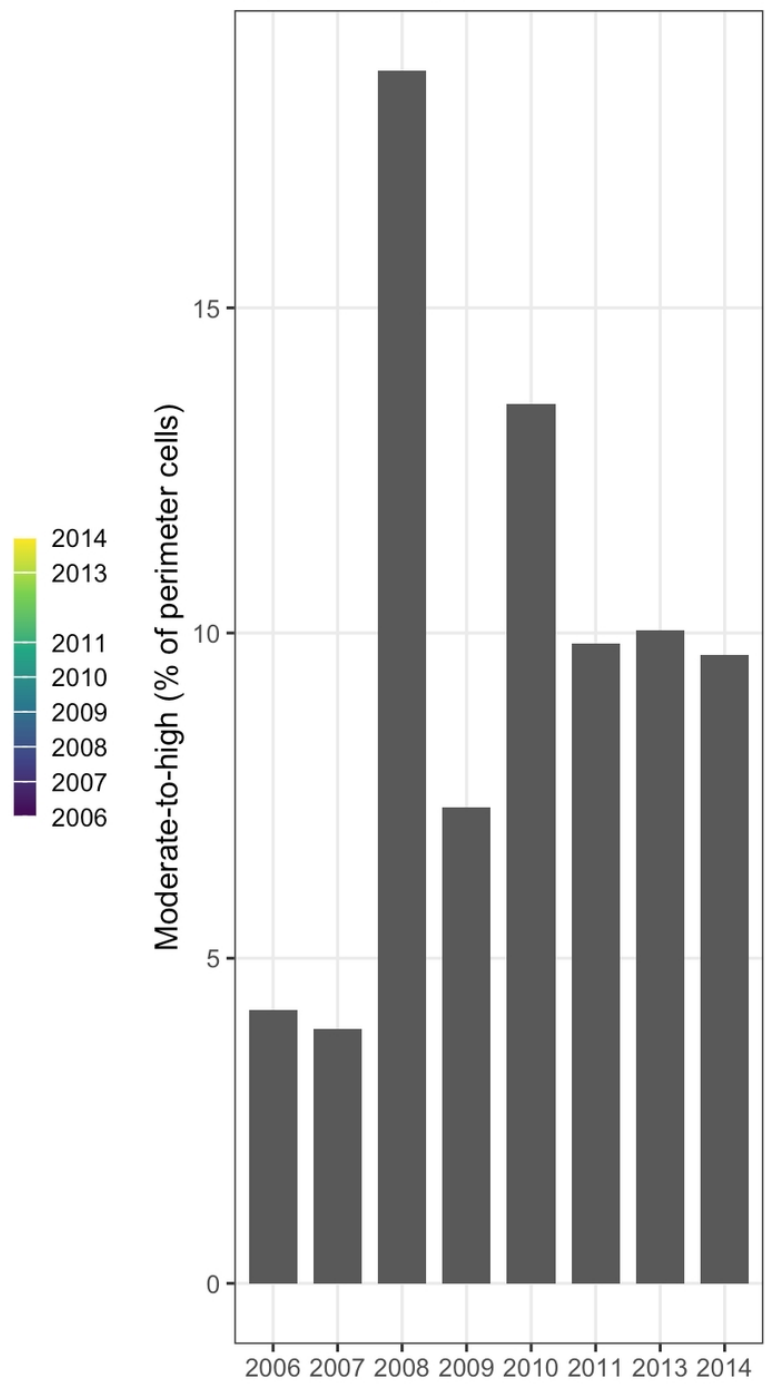
654



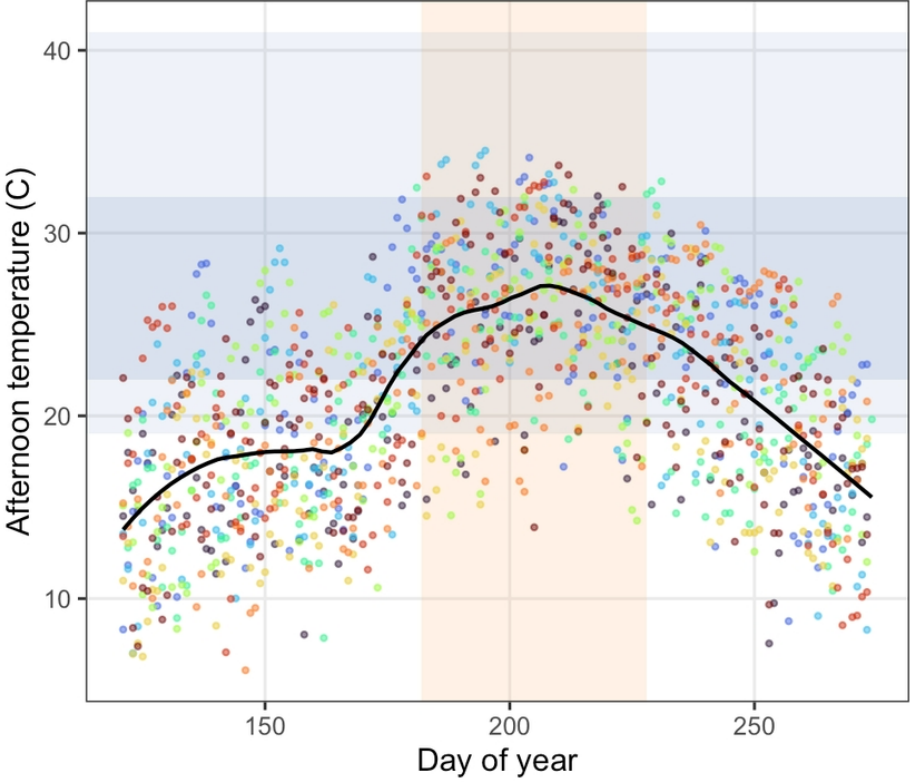
(a) First year classed moderate-to-high



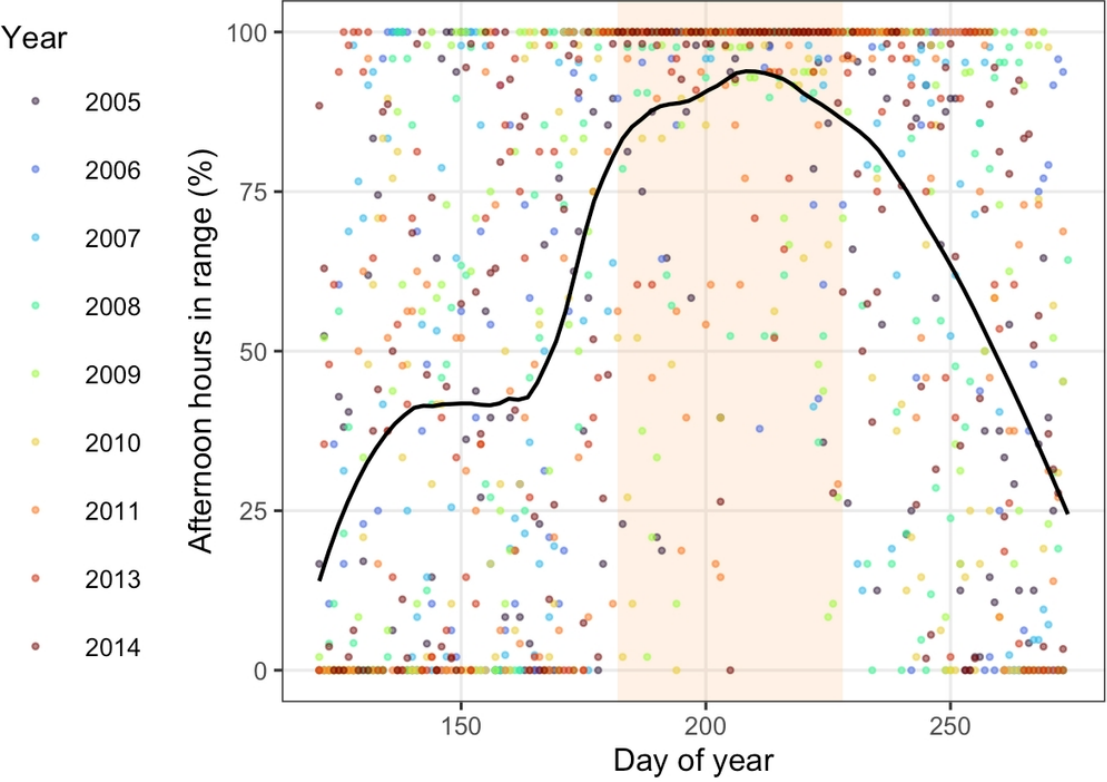
(b) Attack by year



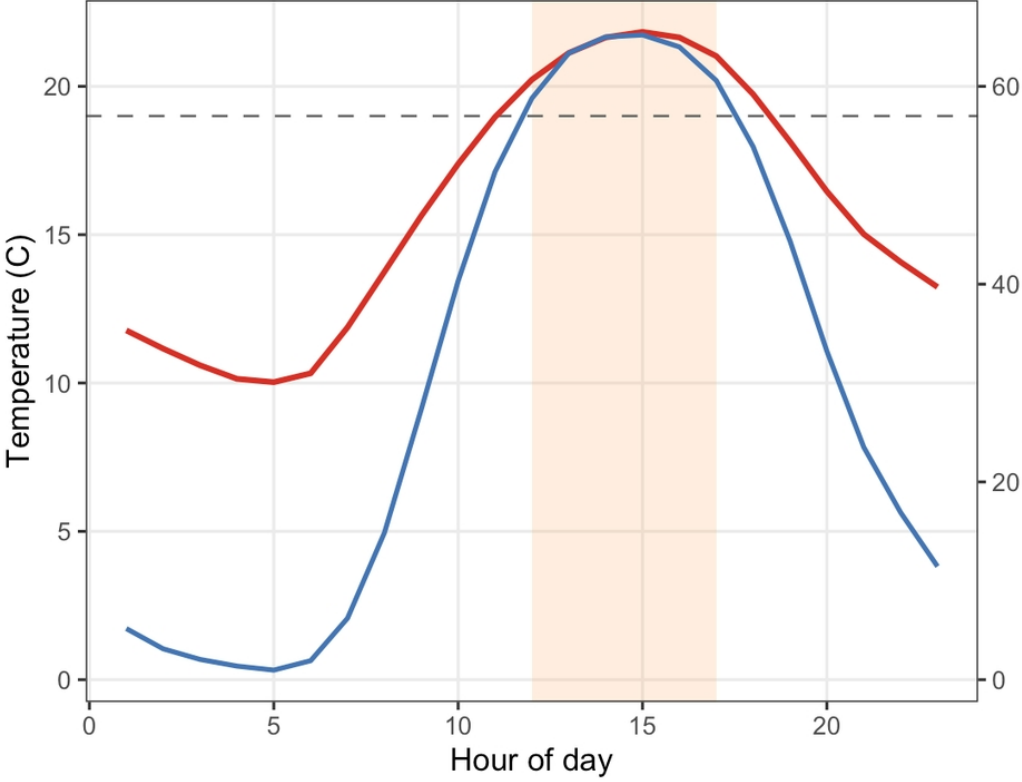
(a) Afternoon temperature and the flight range



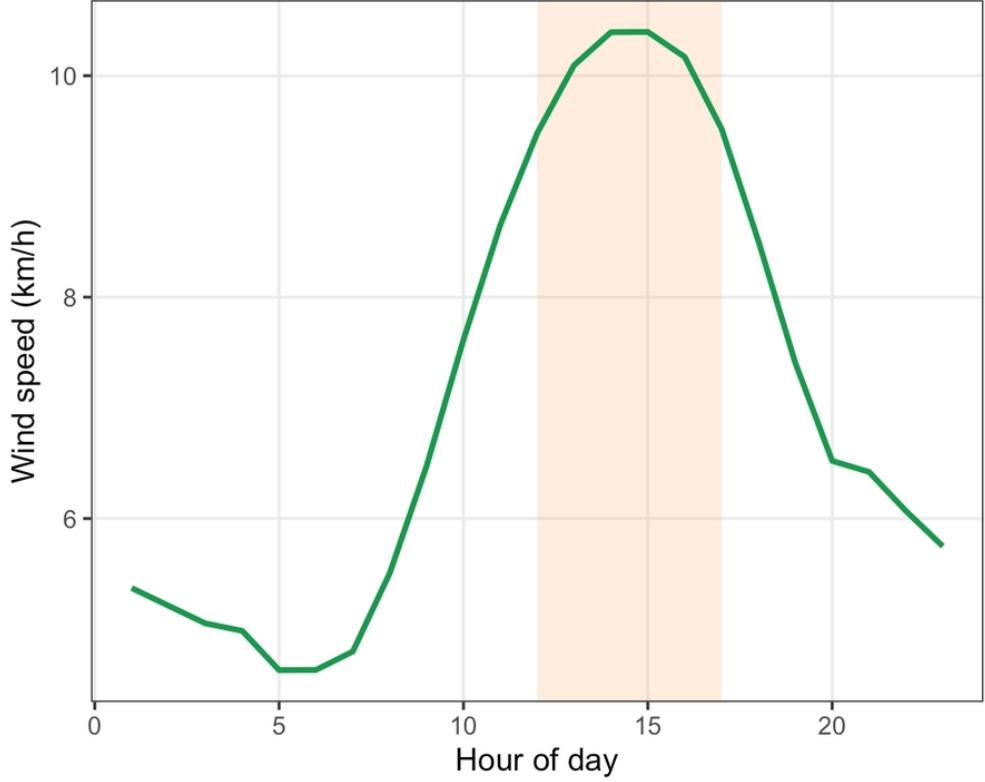
(b) Share of afternoon hours flight is possible

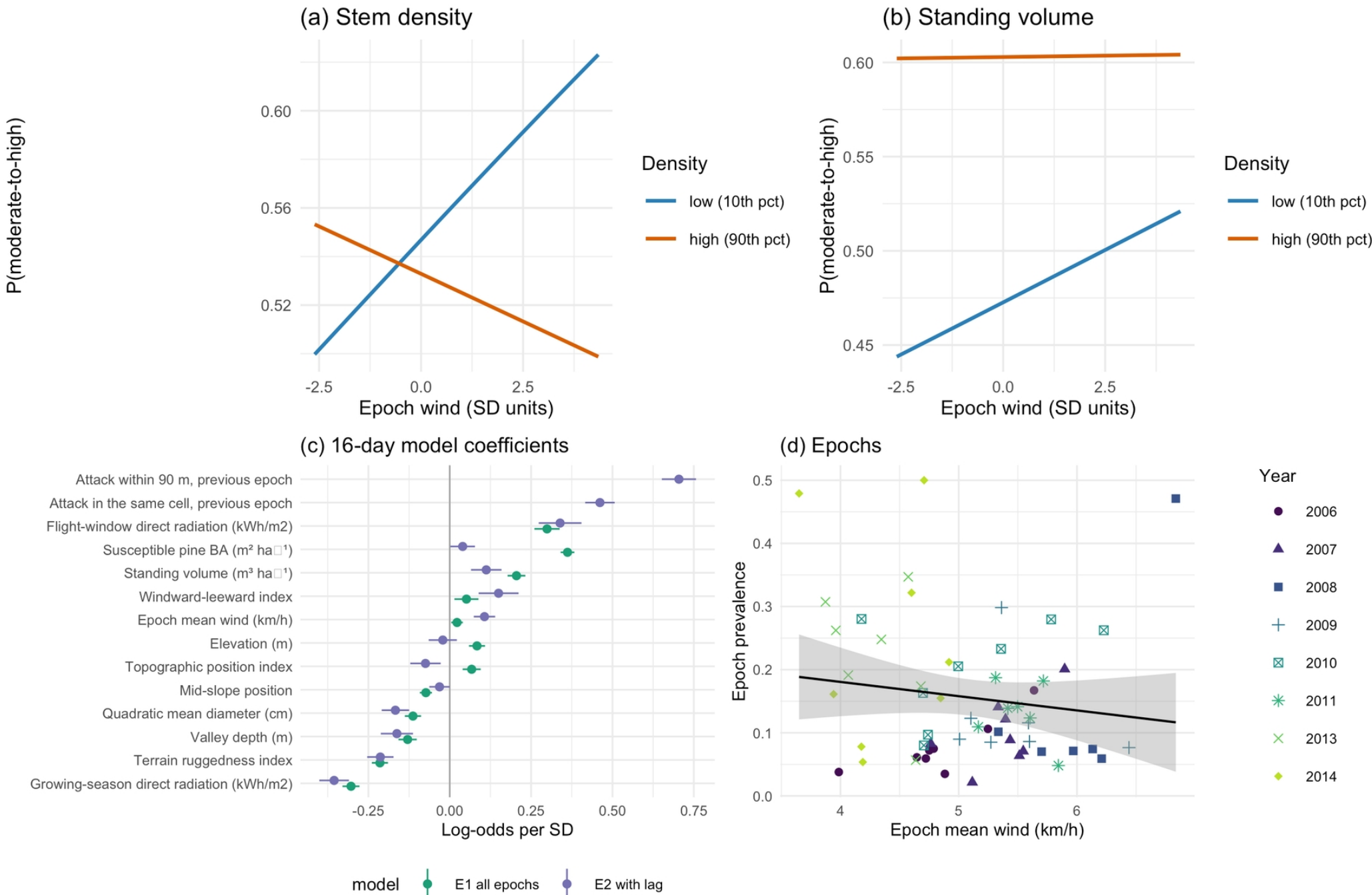


(c) Diurnal cycle of temperature and the flight range



(d) Wind peaks in the same hours





Tables

Table 1

Stand structure across the study perimeter, from the Vegetation Resources Inventory, over 111,707 cell-years. SD was the standard deviation across the landscape and SE the standard error of the mean, which was small at this sample size by construction and measured the precision of the landscape mean rather than of any one cell. Skewness and kurtosis were the bias-corrected third and fourth standardised moments, kurtosis being excess, so that 0 was Gaussian and a positive value heavy-tailed.

Attribute	Mean	SD	SE#	Median	Min	Max	Skew	Kurt.
Stand basal area (m ² ha ⁻¹)	35.56	11.60	0.035	37.39	0.98	62.43	-1.01	1.42
Crown closure (%)	50.10	13.54	0.041	50.00	3.00	70.00	-1.74	3.37
Live stems (n/ha)	772.75	312.80	0.936	775.00	23.00	4600.00	1.70	19.51
Quadratic mean diameter (cm)	27.38	6.40	0.019	26.95	13.55	58.90	0.73	1.24
Stand age (years)	115.46	20.88	0.062	116.00	22.00	237.00	-0.78	1.97
Stand height (m)	26.97	6.27	0.019	27.90	7.00	40.30	-0.26	0.23
Standing volume (m ³ ha ⁻¹)	272.33	131.38	0.393	281.47	0.82	586.61	0.01	-0.35
Lodgepole pine cover (%)	20.11	23.93	0.072	10.00	0.00	100.00	1.45	1.56
Susceptible pine BA (m ² ha ⁻¹)	7.07	8.85	0.026	4.00	0.00	48.30	1.63	2.63

Note the standard error is small because it is computed over 111,707 cell-years, and it measures the precision of the landscape mean, rather than of any one cell.

Table 2

The variables that entered the models after selection, the pathway each served, and the direction expected of it from the mechanism named in the Introduction. AUC was the area under the receiver operating characteristic curve of the univariate fit, with the asterisks beside it marking its significance, * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.

Variable	Pathway	Expected	Univariate AUC
Elevation (m)	Landform	uncertain	0.682****
Susceptible pine BA (m ² ha ⁻¹)	Host size	positive	0.678****
Stand basal area (m ² ha ⁻¹)	Stand density	positive	0.601****
Sky view factor	Topographic shading	positive, as main effect	0.686****

Variable	Pathway	Expected	Univariate AUC
Stand age (years)	Host size	positive	0.571****
July mean wind (km/h)	Flight-window wind, stations	negative	0.563****
Northness	Topographic shading	negative	0.586****
Quadratic mean diameter (cm)	Host size	positive above 25 cm	0.510****
June mean wind (km/h)	Flight-window wind, stations	negative	0.534****
Wind shelter index	Terrain exposure to wind	negative, the index rising windward	0.526****
MicroMet flight-window wind (km/h)	Flight-window wind, terrain-resolved	negative	0.580****
Topographic position index	Terrain shape	more attack in convergent terrain	0.585****
Flight-window direct radiation (kWh/m ²)	Flight-window radiation	positive	0.561****
Convergence index	Terrain shape	more attack in convergent terrain	0.530****
Profile curvature	Terrain shape	more attack in convergent terrain	0.514****

16 Table 3

17 Comparison of the four annual models, each adding one mechanism to the one before. RMSE
18 was the root mean squared error and MAE the mean absolute error, both on the fitted
19 probabilities, and AUC the area under the receiver operating characteristic curve. Brier skill was
20 the improvement over predicting the prevalence for every cell, where 0 was no better than that
21 base rate.

Model	AIC	Δ AIC	RMSE	MAE	AUC	Brier skill
M0 host size, shading, terrain	41,269	1,182	0.396	0.315	0.758	0.166
M1 + stand density	40,971	884	0.394	0.312	0.761	0.175
M2 + terrain, flight radiation	40,649	562	0.393	0.309	0.767	0.182
M3 + interactions	40,087	0	0.390	0.305	0.778	0.193

22 Table 4

23 Coefficients of the full annual model M3, continuous terms ordered by absolute size. Each
24 coefficient was the change in log-odds per standard deviation of its variable, fitted on a class-
25 balanced sample, so the intercept was not the landscape prevalence. SE was the standard error

and z the Wald statistic. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. The geomorphon landform classes of the same model were reported in Table 6.

Term	Beta	SE	z
Stand basal area ($\text{m}^2 \text{ ha}^{-1}$)	+0.430****	0.024	18.20
Northness	+0.384****	0.015	24.85
Elevation (m)	+0.381****	0.016	24.18
Susceptible pine BA ($\text{m}^2 \text{ ha}^{-1}$)	+0.343****	0.015	23.65
Quadratic mean diameter (cm)	-0.341****	0.022	-15.37
Flight-window direct radiation (kWh/m^2)	+0.319****	0.023	13.98
July mean wind (km/h)	+0.286****	0.014	20.95
Sky view factor	+0.285****	0.023	12.60
Wind shelter index	-0.267****	0.021	-12.48
Stand age (years)	+0.201****	0.018	11.31
Convergence index	-0.077****	0.015	-5.07
Stand basal area x July mean wind	-0.077****	0.016	-4.81
Stand basal area x Flight-window direct radiation	+0.073***	0.020	3.58
Profile curvature	-0.059****	0.014	-4.10
June mean wind (km/h)	+0.030*	0.014	2.20
Stand basal area x Wind shelter index	+0.023	0.025	0.89
Topographic position index	-0.015	0.023	-0.64

Table 5

Moderate-to-high beetle disturbance by quadratic mean diameter class, on the balanced sample. QMD was quadratic mean diameter over stems 12.5 cm and up. The 25 cm boundary was the source-sink threshold of the species' bionomics. Intervals were Wilson score intervals on the class proportion, which is why they were asymmetric in the smallest class. Because 30 m cells in a spreading outbreak were not independent, the tests reported in the main text were anti-conservative.

QMD class (cm)	n	Attacked	Attacked (%)	95% CI (%)
<15	395	111	28.1	23.9-32.7
15-20	4,101	783	19.1	17.9-20.3
20-25	9,286	1,998	21.5	20.7-22.4
25-30	17,119	5,391	31.5	30.8-32.2
30-40	10,247	2,291	22.4	21.6-23.2
>40	1,643	217	13.2	11.7-14.9

35 Table 6

36 Geomorphon landform classes in the full annual model M3, as coefficients against the reference
 37 class, slope, the most common class on the perimeter. Cells was the number of perimeter cells
 38 in each class and Attacked the share of those cells classed moderate-to-high over all years. Flat
 39 and pit did not occur on the perimeter, and shoulder, 16 cells with no attack, was merged into
 40 ridge. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$.

Class	Cells	Attacked (%)	Beta	SE
slope (reference)	72,262	9.6		NA
spur	22,050	12.6	-0.075*	0.035
hollow	12,988	6.0	-0.091	0.053
valley	2,963	3.8	-0.676****	0.121
ridge#	1,276	11.0	-1.160****	0.119
peak	168	22.6	-0.523*	0.227

41 # Ridge includes the 16 shoulder cells on the perimeter. No shoulder cell was classed as attacked in any year, so the class could
 42 not be estimated on its own and was counted with ridge, its neighbour on the geomorphon scale.

Supporting Information

Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*). Terrain, stand density and flight-period wind as controls on attack in the Selkirk Mountains of British Columbia

Seamus Murphy

Table S1

The datasets this study combined, with the structure and resolution of each. Spatial resolution was the grid on which a variable was analysed and temporal resolution the interval at which it varied. The response varied every 16 days, the inventory once a year, the station wind every hour and the terrain not at all, which was the unevenness the wind analysis depended on.

Dataset	Source	Variables	Spatial	Temporal	Period	n
Beetle attack	Landsat 5 and 8 Collection 2 Level-2	Moderate-to-high NDMI	30 m	1 year	2006-2014	8 years
Beetle attack	Landsat 5 and 8 Collection 2 Level-2	Moderate-to-high NDMI	30 m	16 days	2006-2014	59 epochs
Stand structure	VRI Historical, BC Data Catalogue	BA, volume, stems, quadratic mean diameter, age, height	30 m (raster)	1 year, nonoriented	2005-2014	9 windows
Terrain	NRCan High-Res DEM, SAGA indices	Geomorphons (incl. radiation, exposure, landform)	30 m	Static	n/a	8 fitted
Station wind	Env. & Climate Change Canada	Speed and direction	4 to 7 station	1 hour	2005-2014	236,079 hourly
Terrain-resolved Model	DEM-conditioned MicroMet wind field of station data	Weighting factor, modified speed, diverted direction	30 m	16 days, & 1 year	2005-2014	16 sectors (22.5°)
Model frame	Rows joined above, (one row / cell-year)	Response & covariates, one row per cell-year	30 m	1 year	2006-2014	111,707 cell-years
Model frame / 16-	Rows joined above per Landsat pass (one row / cell-	Response & covariates, one row per cell-epoch	30 m	16 days	2006-2014	66,302 cell-

Table S2

Moderate-to-high beetle disturbance by year inside the study perimeter.

Year	Cells	Moderate-to-high (%)
2006	13,224	4.2
2007	13,225	3.9
2008	13,285	18.6
2009	13,556	7.3
2010	13,538	13.5

Year	Cells	Moderate-to-high (%)
2011	14,944	9.8
2013	14,943	10.0
2014	14,992	9.7

13 Methods S1

14 The wind component of the MicroMet model of Liston & Elder (2006) is seven equations,
15 implemented directly from the source paper with its equation numbers given below.

16 Terrain slope β and slope azimuth ξ came from the elevation model, equations 12 and 13 of Liston
17 & Elder (2006). Curvature Ω_c is a cell's elevation minus the mean of the two opposite cells one
18 curvature length scale away, taken on four direction lines and averaged, their equation 14, and
19 the length scale was estimated as the first lag at which elevation autocorrelation fell below 0.5,
20 which was 600 m here. Assigning θ to wind bearing, slope in the wind direction was computed as
21 follows

$$22 \quad \Omega_s = \beta \cos(\theta - \xi) \quad (1)$$

23 with Ω_s and Ω_c each scaled to $[-0.5, 0.5]$. The terrain weighting factor applied to the observed
24 speed W was derived using

$$25 \quad W_w = 1 + \gamma_s \Omega_s + \gamma_c \Omega_c, \quad \gamma_s = \gamma_c = 0.5 \quad (2)$$

26 This provided the basis for calculation of the terrain-modified speed using

$$27 \quad W_t = W_w W \quad (3)$$

28 and a diversion of wind direction computed as

$$29 \quad \theta_d = -1/2 \Omega_s \sin[2(\xi - \theta)]. \quad (4)$$

30 By [Equation 2](#), W_w depends on direction and not on speed, so it was computed once for each of
31 16 wind direction sectors of 22.5 degrees, the sixteen points of the compass, and each hourly
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Elevation (m)	Landform	0.682****	retained	0.336	yes
Susceptible pine BA ($\text{m}^2 \text{ha}^{-1}$)	Host size	0.678****	retained	0.427	yes
Stand basal area ($\text{m}^2 \text{ha}^{-1}$)	Stand density	0.601****	retained	0.347	yes
Sky view factor	Topographic shading	0.686****	retained	0.315	yes
Stand age (years)	Host size	0.571****	retained	0.068	
July mean wind (km/h)	Flight-window wind, stations	0.563****	retained	0.249	yes
Northness	Topographic shading	0.586****	retained	0.314	
Quadratic mean diameter (cm)	Host size	0.510****	retained	-0.124	
June mean wind (km/h)	Flight-window wind, stations	0.534****	retained	-0.025	
Wind shelter index	Terrain exposure to wind	0.526****	retained	-0.205	yes
MicroMet flight-window wind (km/h)	Flight-window wind, terrain-resolved	0.580****	retained	0.253	yes
Topographic position index	Terrain shape	0.585****	retained	-0.168	yes
Flight-window direct radiation (kWh/m^2)	Flight-window radiation	0.561****	retained	0.206	yes
Convergence index	Terrain shape	0.530****	retained	-0.016	
Profile curvature	Terrain shape	0.514****	retained	-0.010	
Height above valley floor (m)	Landform	0.646****	penalty	0.000	
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Crown closure (%)	Stand density	0.524****	penalty	0.000	
Vector ruggedness measure	Terrain shape	0.600****	penalty	0.000	
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Normalised height	Landform	0.584****	inflation	NA	
Flight-window mean wind (km/h)	Flight-window wind, stations	0.558****	inflation	NA	
Eastness	Topographic shading	0.549****	inflation	NA	
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Candidate	Pathway	Univariate AUC	Stage	Lasso coefficient	Protected
Lodgepole pine cover (%)	Host size	0.656****	collinearity	NA	
Slope (degrees)	Terrain shape	0.645****	collinearity	NA	
Positive openness	Terrain exposure to wind	0.636****	collinearity	NA	
Wind exposition index	Terrain exposure to wind	0.563****	collinearity	NA	
Standing volume ($\text{m}^3 \text{ ha}^{-1}$)	Stand density	0.555****	collinearity	NA	
Flight-window windy hours (share above 15 km/h)	Flight-window wind, stations	0.548****	collinearity	NA	
Flight-window calm hours (share below 5 km/h)	Flight-window wind, stations	0.547****	collinearity	NA	
Multi-scale topographic position	Terrain shape	0.540****	collinearity	NA	
August mean wind (km/h)	Flight-window wind, stations	0.534****	collinearity	NA	
Plan curvature	Terrain shape	0.528****	collinearity	NA	
solar_flight_diffuse	Flight-window radiation	0.528****	collinearity	NA	
Windward-leeward index	Terrain exposure to wind	0.522****	collinearity	NA	
Growing-season total radiation (kWh/m^2)	Topographic shading	0.517****	collinearity	NA	
Growing-season direct radiation (kWh/m^2)	Topographic shading	0.516****	collinearity	NA	
Flight-window 95th percentile wind (km/h)	Flight-window wind, stations	0.527	univariate	NA	
Stand height (m)	Host size	0.507	univariate	NA	
Topographic wetness index	Landform	0.506*	univariate	NA	
Negative openness	Terrain exposure to wind	0.501	univariate	NA	
Heat load index	Topographic shading	0.499	univariate	NA	

Table S4

The wind variable of the sixteen-day models was the epoch wind regime, the mean over every hourly station observation in the sixteen days an epoch covered, each hour modified over the terrain by its own bearing. It was not restricted to the flight window, and that choice followed from the way the insect was known to disperse. Jackson et al. (2008) tracked mountain pine beetle on weather radar and confirmed the returns by aerial capture, finding beetles “at altitudes up to more than 800 m above the forest canopy” and estimating that those in flight above the canopy “may move 30 to 110 $\text{km} \cdot \text{day}^{-1}$ ”, at a mean density of 4,950 and a maximum of 18,600 beetles per hectare. Ainslie & Jackson (2010) subsequently reproduced that above-canopy dispersion with an atmospheric model. An insect carried in the boundary layer at those heights and over those distances was exposed to the atmospheric regime of the period rather than to the wind measured within a stand between noon and five, and the variable was named accordingly.

The 16-day model was refitted under four definitions of the window (Table S4), every hour, the 12:00 to 17:00 hours the flight peak occupied, the 11:00 to 18:00 hours that fell inside the thermal limits on this landscape more than half the time, and only those hours whose observed temperature fell inside the 19 to 41 degrees C range itself. The stem-density interaction was negative under all four definitions and distinguishable from zero under three, at -0.049 over the full regime and -0.037 and -0.041 over the two clock windows, and fell to -0.011 ($P = 0.173$) when the hours were restricted to those inside the thermal limits, where the standing-volume interaction was instead the larger of the two at -0.040 ($P < 0.001$). One of the two density terms therefore carried the interaction under every definition, but which one depended on how the window was drawn, so the mechanism did not rest on the choice while its expression through stems or through volume did.

The 16-day model refitted under four definitions of the wind window. The epoch wind regime, every hour, was the definition used throughout the article. Coefficients were log-odds per standard deviation and AUC the area under the receiver operating characteristic curve. Significance was marked * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$. The thermal-limits row was fitted on fewer cell-epochs because two epochs had too few qualifying hours, so its row was not directly comparable with the other three.

Definition	Cell-epochs	Wind (km/h)	Stems x wind	Volume x wind	AUC
Epoch wind regime (all hours)	66,302	5.01	-0.0491****	-0.0169*	0.673
12:00-17:00	66,302	6.41	-0.0369****	-0.0131	0.673
11:00-18:00	66,302	6.28	-0.0413****	-0.0173*	0.673
Inside the 19-41 C range	64,895	5.64	-0.0108	-0.0397****	0.671

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